

Stochastic casualty response planning with multiple classes of patients

Pedram Farghadani-Chaharsooghi ^{a,b} , Hossein Hashemi Doulabi ^{a,b} , Walter Rei ^{c,b} , Michel Gendreau ^{d,b}

^a Department of Mechanical, Industrial and Aerospace Engineering, Concordia University, Montréal H3G 1M8, Québec, Canada

^b CIRRELT, Université de Montréal, Montréal H3C 3J7, Québec, Canada

^c Département d'Analytique, Opérations et Technologies de l'Information, Université de Québec à Montréal, Montréal H2X 3X2, Québec, Canada

^d Department of Mathematics and Industrial Engineering, Polytechnique Montréal, Montréal H3C 3A7, Québec, Canada

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ABSTRACT

In this paper, we study the stochastic casualty response planning problem (CRP) in the context of providing treatments to multiple classes of patients with different types of injuries. In this general setting, both patients' demands and hospitals' treatment capacity are considered uncertain. To the best of our knowledge, this is the first time that this problem is solved. We propose a novel two-stage stochastic mixed-integer programming model which, in the first stage, determines the location of the Alternative Care Facilities (ACFs) and allocates different resources, such as rescue vehicles, medical equipment, and physicians, to them. In the second stage, this model helps decide how to allocate patients with multiple injuries to either ACFs or hospitals, considering their care itineraries and available resources. Moreover, it recommends potential patient transfers between ACFs and hospitals when required. Furthermore, we introduce an alternative two-stage stochastic model that is more compact than the first. This formulation significantly reduces solution times. We also provide an equivalency proof between the two formulations. As the solution method, we develop both the L-shaped algorithm, a pure cutting-plane method tailored to our stochastic mathematical model, and the branch-and-Benders-cut (B&BC) algorithm. To further enhance the efficiency of these algorithms, we develop a wide range of acceleration techniques, including Benders dual decomposition, Lagrangian dual decomposition, a multi-cut reformulation, Pareto-optimal cuts, and the inclusion of lower bounding functional valid inequalities. We carry out extensive computational experiments demonstrating that these algorithmic enhancements dramatically improve the performance of the B&BC algorithm, reducing the average optimality gap from 7898% in the standard B&BC algorithm to just 0.92% in the enhanced version. Additionally, we benchmark our approach against the progressive hedging algorithm (PHA), a widely used decomposition method in disaster response operations, to further assess its effectiveness. Finally, we present a case study from the 2011 Van earthquake in Turkey, demonstrating the applicability and efficiency of our optimization methods.

1. Introduction

Catastrophic health events (CHEs) are natural or man-made disasters that result in large numbers of injured individuals and put healthcare systems under pressure to take emergency actions. After CHEs, injured people require immediate medical attention, and their life often depends on a quick reaction from emergency healthcare providers. As a result, efficient resource management in the initial few hours following a disaster is crucial to enhance the casualty survival rate. Casualty response planning refers to the process of collaboration between organizations to efficiently manage limited resources to help survivors in affected areas and transfer them to hospitals. There have been numerous devastating natural disasters in the past few decades, which have motivated many researchers to work on casualty response planning problems. Notably, Turkey's 2023 earthquake claimed 59,259 lives, Haiti's 2010 earthquake resulted in a staggering 250,000 deaths, and the 2004 South Asia tsunami took 227,898 lives, marking these events as some of history's most devastating natural disasters.

* Correspondence to: Department of Mechanical, Industrial and Aerospace Engineering, Concordia University, 1455 De Maisonneuve Blvd. W., Montréal H3G 1M8, Québec, Canada.

E-mail addresses: pedram.farghadani-chaharsooghi@mail.concordia.ca (P. Farghadani-Chaharsooghi), hossein.hashemi@concordia.ca (H. Hashemi Doulabi), rei.walter@uqam.ca (W. Rei), michel.gendreau@polymtl.ca (M. Gendreau).

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In the context of CRP, using Alternative Care Facilities (ACFs) is one of the most common ways to diagnose an overwhelming number of casualties and treat them after a disaster. ACFs are set up as temporary healthcare facilities with the primary objective of delivering essential care during times of heightened demand, such as in the aftermath of a natural disaster. Stadiums, parks, universities, and schools are some instances of locations where ACFs can be established (Caunhye and Nie, 2018). The advantages of using ACFs will not realize without a suitable location-allocation plan. For example, during the 1995 Oklahoma City bombing, emergency medical personnel set up an ACF in the form of a triage center. However, they could not assess almost half of the victims because the triage center was initially located too far from the assault site (Larson et al., 2006).

The literature on CRP can be categorized into three main categories: deterministic, stochastic, or robust approaches. Deterministic models, while being widely used, have a notable drawback. They may yield infeasible plans or lead to unexpected penalty costs due to ignoring the uncertainty of demands and hospitals' treatment capacity. Stochastic models address these limitations by incorporating uncertainty into the planning through scenarios, acknowledging that the full scope of disaster needs becomes clear over time.

Such needs are addressed by first-stage and recourse decisions in the stochastic optimization paradigm. These decisions typically evaluate needs using a combination of pre- and post-disaster data, including governmental records, technology outputs like satellite imagery, and on-the-ground reports from key informants, such as community leaders (Grass et al., 2023). Key informants play a crucial role in estimating casualty demand by providing informed assessments based on their expertise, observations, and available data sources. These key informants, who may include medical experts, emergency responders, and disaster assessment teams, do not attempt to predict specific injuries but rather classify casualties into broad patient categories requiring different treatment types. This classification allows for a structured medical resource allocation strategy that aligns with expected treatment needs. Since different disaster types result in varying casualty distributions, practitioners rely on structured scenario-generation processes that incorporate multiple sources of information to construct a realistic representation of potential demand.

In real-world disaster response planning, scenario generation typically relies on historical data, field assessments, and expert evaluations. Historical data provides baseline distributions of casualties from past events, but given the uniqueness of each disaster, updated assessments become crucial. Ground teams and emergency response personnel commonly refine these estimates using various assessment methods, including aerial observations, satellite imagery, and direct communication with community-based informants (Clingerman, 2019). In practice, assessments often involve multiple key informants from the affected communities who independently provide estimates of casualties, injury types, and the severity of impact. For example, in the aftermath of the 2004 tsunami, assessment teams in Indonesia relied on information from multiple informants within communities to verify and update casualty and damage estimates (Clingerman, 2019). Given that no single informant's evaluation is likely to be entirely accurate, decision-makers aggregate these independent assessments to generate a representative range of plausible scenarios. Consequently, the scenarios used in stochastic optimization reflect combined expert perspectives, historical trends, and observed real-world conditions, ensuring robustness in disaster planning decisions.

Practitioners who handle casualty response planning often work with scenarios in this way. In some cases, they simply rely on deterministic models using average estimates from multiple sources, which may lead to overly rigid plans that do not account for the true variability of demand. While averaging assessments can provide a simplified representation, it inherently discards crucial information about uncertainty, potentially leading to inefficient or infeasible response strategies. Stochastic optimization preserves this uncertainty by explicitly incorporating scenario variations, allowing decision-makers to develop more resilient and adaptive plans. Instead of committing to a single forecast, a stochastic model ensures that resource allocation decisions remain flexible and capable of adapting to different demand realizations.

Another approach to addressing uncertainty is robust optimization, which focuses on finding the best possible decision under the worst-case scenario. Choosing between stochastic optimization and robust optimization depends largely on the confidence level in the available data. If field assessments provide sufficient and reliable information, stochastic optimization offers a balanced approach that accounts for a range of possible outcomes. However, when field data is scarce or unreliable, robust optimization provides a more conservative planning strategy that prepares for the most extreme scenarios. Given our assumption that key informants provide credible assessments backed by multiple sources, we adopt a scenario-based two-stage stochastic approach to optimize decision-making under uncertainty. This methodology ensures that casualty response planning remains realistic, data-driven, and adaptable to the complexities of post-disaster environments. While we briefly review the major works in this area, we refer interested readers to the comprehensive surveys by Farahani et al. (2020) and Dönmez et al. (2021) for a more in-depth literature review.

In the first category, many researchers assumed that all parameters are deterministic and given (e.g., see Apte et al., 2015; Caunhye et al., 2015; Liu et al., 2019; Setiawan et al., 2019). Apte et al. (2015) developed a mixed-integer programming model to identify the best locations for casualty collection points and to allocate vehicles, resources, and injured people to them, in order to maximize the number of survivors. Caunhye et al. (2015) offered a location-allocation optimization model to minimize the total transportation time, which simultaneously locates ACFs, prioritizes casualties, and considers self-evacuees effects. Setiawan et al. (2019) formulated three mathematical programming models to take into account different levels of coordination for the distribution of relief resources and the allocation of vehicles to minimize the number of victims. Liu et al. (2019) considered ambulances and helicopters to move casualties. Their model determines the optimal locations for temporary healthcare services and recommends the allocation of medical services to injured people.

On the other hand, many researchers in the literature have proposed stochastic optimization (Caunhye and Nie, 2018; Alizadeh et al., 2019; Oksuz and Satoglu, 2020; Chang et al., 2023) and robust optimization models (Hu et al., 2019a; Li et al., 2020; Sun et al., 2021; Yin et al., 2024) to deal with uncertainty in catastrophic health events. Caunhye and Nie (2018) proposed a three-stage stochastic programming model which determines the location of ACFs in the first stage by considering self-evacuees' movement. Then, in the second and third stages, the model addresses the assignment of casualties to triage and treatment, respectively. Alizadeh et al. (2019) proposed a two-stage robust stochastic optimization model with uncertain levels of injuries. They demonstrated that incorporating uncertainty into strategic-level decision-making leads to better design outcomes for the casualty collection points. Oksuz and Satoglu (2020) developed a two-stage stochastic programming model for locating ACFs in the event of an earthquake. Their stochastic model aimed to minimize establishing ACFs and casualties' transportation costs. They incorporated innovative elements into their model by accounting for triage and α -reliability constraints. α -reliability constraints are employed to guarantee a predefined level of reliability (α) within the solutions. The reliability concept ensures that the solution meets all demands in scenarios that encompass at least 100α percent of all possible outcomes. Most recently, Chang et al. (2023) developed a strategy to determine the locations of ACFs and efficiently allocate the limited resources of emergency medical services (EMS) for quick casualty transportation to suitable hospitals, thereby enhancing casualty survival rates. They introduced a novel two-stage simulation-optimization algorithm to tackle the ACF location and EMS resource allocation problem, which combines binary and integer variables. The primary objective was to minimize the expected time for complete casualty delivery from the disaster area to hospitals.

Table 1
An overview of latest related works.

Authors	FL	RA	VA	CA	PSA	ML-Inj	MI-PC	HV	Uncertainty type	UA	Objective function	Solution approach
Apte et al. (2015)	✓	✓	✓	✓					Deterministic model	N/A	Minimizing transportation time	MIP
Caunhye et al. (2015)	✓			✓		✓			Deterministic model	N/A	Minimizing transportation time	MIP
Caunhye and Nie (2018)	✓			✓		✓			Number of casualties Proportion of self-evacuees Transportation times	SP	Minimizing transportation time	Benders decomposition
Liu et al. (2019)	✓			✓		✓			Deterministic model	N/A	Maximizing expected survivals Minimizing total operational cost	ϵ -constraint method
Setiawan et al. (2019)	✓	✓	✓	✓				✓	Deterministic model	N/A	Minimizing unmet demand and victim suffering	Heuristic algorithm
Alizadeh et al. (2019)	✓			✓		✓			Number of casualties Transportation capacity	HRS	Minimizing transportation cost and mortality	Sample average approximation
Hu et al. (2019a)	✓			✓					Disaster occurrence time	RO	Minimizing total operational cost and maximum travel distance	Benders decomposition
Li et al. (2020)	✓	✓		✓					Disaster severity Transportation time Number of casualties	HRS	Minimizing setup, operational and evacuation costs	Progressive hedging algorithm
Oksuz and Satoglu (2020)	✓			✓		✓			Earthquake magnitude	SP	Minimizing setup and transportation costs	MIP
Sun et al. (2021)	✓	✓		✓		✓			Number of casualties Number of rescue supplies Transportation time	RO	Minimizing injury severity score and total operational cost	ϵ -constraint method
Chang et al. (2023)	✓		✓						Triage and treatment time Transportation time	SP	Minimizing expected casualty delivery time	Simulation optimization
Yin et al. (2024)	✓	✓		✓					Number of casualties	RO	Minimizing setup, transportation and inventory costs	Benders decomposition
This study	✓	✓	✓	✓	✓	✓	✓	✓	Number of casualties Hospital treatment capacity	SP	Minimizing setup, transportation, operational, and unmet demand costs	Accelerated B&BC algorithm

FL: Facility location; RA: Resource allocation; VA: Vehicle allocation; CA: Casualty allocation; PSA: Physician specialty assignment; ML-Inj: Multi-level injuries; MI-PC: Multiple injuries (patient classes); HV: Heterogeneous vehicle; UA: Uncertainty approach; N/A: Not applicable; SP: Stochastic programming; HRS: Hybrid robust and stochastic method; RO: Robust optimization; MIP: Mixed-integer programming model without a specific solution approach.

In the context of robust optimization, [Hu et al. \(2019a\)](#) paid attention to diurnal population shifts in urban areas and developed a multi-objective robust optimization model. They proposed an accelerated Benders decomposition (BD) algorithm to solve large-size problems. While all previous works covered only the primary disaster, [Li et al. \(2020\)](#) took secondary shocks into account by offering a scenario-based robust programming model. It is worth mentioning that they assumed all patients have the same medical needs and belong to the same category. However, CRPs are significantly more complex since patients have multiple types of injuries resulting in different categories of patients with needs for different surgical specialties. [Sun et al. \(2021\)](#) introduced a bi-objective robust optimization model for strategic and operational response within a three-level rescue chain comprising casualty clusters, temporary facilities, and general hospitals. Their approach incorporated uncertainties related to relief supply distribution, demands, and transportation times. To address this model, they used Lagrangian relaxation and ϵ -constraint techniques. [Yin et al. \(2024\)](#) presented a distributionally robust model for complex disaster response, involving facility location, supply inventory, and evacuation planning. Their solution utilized a branch-and-Benders-cut (B&BC) algorithm.

A summary of the CRP literature and the contributions of this study are presented in [Table 1](#). As shown, our work goes beyond integrating multiple well-established elements, such as facility location, resource allocation, casualty allocation, and the use of heterogeneous rescue vehicles. A key novelty of our study is the introduction of care itineraries through the consideration of multiple injuries, i.e., different patient classes. By incorporating care itineraries, we recognize the need for physician specialty assignment—another aspect that, to our knowledge, has not been explicitly addressed in previous CRP models. These contributions enhance the realism of post-disaster planning by accounting for diverse medical needs while also introducing an additional combinatorial dimension to the problem.

Various injuries after CHEs can significantly impact the health and survival of individuals affected by the disaster. For example, earthquakes can cause a variety of injuries, including fractures, lacerations, nerve injuries, and burns ([Del Papa et al., 2019](#)). These injuries can be life-threatening and may require immediate medical attention. Additionally, the combination of different injuries can increase the risk of infection, blood loss, and other complications. [Lu-Ping et al. \(2012\)](#) reported that after the Wenchuan earthquake in 2008, 45.8%, 9%, and 1.5% of injured people had two, three, and more than four injuries, respectively. [Del Papa et al. \(2019\)](#) found that 52% of people injured in the 2009 L'Aquila earthquake had multiple injuries. In the 2013 Lushan earthquake in China, this number was 55.7% ([Kang et al., 2015](#)). Although patients commonly sustain multiple injuries following disasters, the studies previously reviewed do not consider patients with multiple injuries post-disaster. This is an important point to consider in casualty response planning since it can significantly affect the allocation of medical resources and patients to ACFs. Ignoring the variety of injuries and classifying patients only as a high or low priority can result in sending them to ACFs that are not appropriately equipped for any of their specific injuries.

In addition to multiple injuries, there are several important operational details that have yet to be addressed in the literature. For example, different patients have different needs for renewable and non-renewable resources. Therefore, we have to plan ahead for assigning physicians

to ACFs accordingly. For instance, we cannot send a patient with a fracture injury to an ACF that does not have an orthopedic doctor. On the other hand, non-renewable resources such as medical supplies and drugs also play an important role in the functioning of hospitals and ACFs. For example, Kirschner wires (K-wires) are small, thin, metal pins commonly used in orthopedic surgeries. However, their demand may spike during an emergency, and it is impossible to perform orthopedic surgery without them. This highlights the importance of proper resource allocation in CRPs. Additionally, rescue vehicles are crucial for evacuating people from hazardous locations and transporting emergency personnel and supplies. A suitable rescue vehicle allocation after a disaster helps to ensure that emergency medical services personnel and supplies reach the affected population as quickly and efficiently as possible. It can also improve the effectiveness of the emergency response and help save more lives.

Although there is a rich literature on CRPs, to the best of our knowledge, no paper has addressed patients with multiple types of injuries and the above mentioned features simultaneously. The main contributions of this research are as follows.

- For the first time, we investigate the stochastic casualty response planning problem concerning treating multiple patient classes with varying injury types. Our approach also involves assigning the appropriate type of physicians, non-renewable resources, and rescue vehicles to ACFs based on each patient's specific injuries.
- We propose a primary two-stage stochastic mixed-integer programming model to formulate this problem. Then, we offer an alternative two-stage stochastic formulation, which is more compact than the first. As a result, the second model is significantly more efficient. We also present a proof establishing the equivalence between the two formulations.
- We develop an L-shaped algorithm and a B&BC algorithm to solve our more efficient stochastic programming model. We significantly improve the algorithms by using the Benders dual decomposition (BDD), Lagrangian dual decomposition (LDD), a multi-cut reformulation, Pareto-optimal (PO) cuts, and the inclusion of lower bounding functional valid inequalities (LBF). These enhancements improve the overall performance of the proposed algorithms.
- We present extensive computational results for the CRP when dealing with multiple injuries for each patient. In our computational experiments, we compare the proposed two-stage stochastic programming models and evaluate the performance of the proposed decomposition algorithms and the effect of the developed enhancements.
- To assess the effectiveness of our proposed approach, we benchmark it against the PHA, a widely used decomposition method in disaster response optimization, ensuring a meaningful comparison in terms of solution quality and computational efficiency.
- We also evaluate the efficiency of our best model and algorithm in an earthquake case study from Turkey.

The structure of this paper is organized as follows. In Section 2, we describe the problem setting. Then, we present the original and improved two-stage stochastic programming models in Sections 3 and 4, respectively. Section 5 details the proposed L-shaped algorithm and explains the various enhancement techniques we use to improve the algorithm's performance. Our extensive computational results on the CRP are presented in Section 6. In Section 7, we present a case study and evaluate out-of-sample performance, as well as the impact of hospital capacity uncertainty. Finally, we draw conclusions and discuss future directions in Section 8.

2. Problem description

Generally, disaster management includes four main phases: mitigation, preparedness, response, and recovery. Mitigation involves understanding vulnerabilities to potential hazards and implementing protective measures to reduce risk and increase resilience. Preparedness involves evaluating plans to save lives and coordinate response efforts before a disaster strikes. The primary goal is to achieve an adequate state of preparedness for emergency response through initiatives that strengthen the technical and managerial capabilities of governments, organizations, and communities. The response phase aims to provide immediate assistance to victims, distribute relief supplies, and evacuate affected populations to safe areas. The recovery phase occurs after the emergency and focuses primarily on activities such as debris removal, rebuilding damaged structures, and restoring critical infrastructure. Mitigation and preparedness occur before a tragedy, while response and recovery occur thereafter (Farahani et al., 2020).

The problem that we study in this paper deals with the response phase, which occurs immediately post-disaster. These problems are generally approached through static modeling, which optimizes location and resource allocation decisions while incorporating uncertainties (Caunhye and Nie, 2018; Liu et al., 2023). Static models provide solutions for a specific point in time, making them well-suited for rapid decision-making in post-disaster scenarios (Dalal and Üster, 2018; Bayram and Yaman, 2018; Wang and Nie, 2022). For example, Dalal and Üster (2018) explored a static humanitarian response network that integrates relief supply and evacuation, solving it using a BD algorithm. Similarly, Bayram and Yaman (2018) developed a two-stage stochastic static evacuation planning model that accounts for uncertainty in both demand and travel time. Their approach optimally determines shelter locations and assigns evacuees while ensuring that travel distances remain within an acceptable limit. In another study, Wang and Nie (2022) examined a static location-allocation problem that incorporates traffic congestion effects into relief supply planning, formulating a two-stage mixed-integer stochastic programming model and solving it with a generalized BD algorithm. While some studies have explored dynamic approaches that account for evolving demand and uncertainty over time, static models remain widely used for their ability to provide effective solutions in time-sensitive disaster response settings.

In our study, we adopt a static approach to model the post-disaster response phase, ensuring effective medical care supply planning within a predetermined time horizon. The time horizon for the decision-making environment begins once the CHEs commence and continues until all affected individuals have been successfully evacuated. The nature of the disaster determines the duration of this time frame. Our study assumes that the critical period, which includes the time during which victims are transported to hospitals or ACFs, is constrained within this predetermined time horizon, which is set at 7 days. Therefore, our primary objective is to design the appropriate medical care supply to meet the overall demand observed during the post-disaster response phase. In this problem, we make three important decisions after a CHE that includes:

1. Determining the location of ACFs,
2. Allocating renewable and non-renewable medical resources such as physicians, rescue vehicles, medications, and medical equipment to ACFs, and
3. Transporting casualties to ACFs and hospitals.

The first two sets of decisions are made immediately after the disaster based on the initial information received about the intensity and geographical dispersion of the disaster. Therefore, we have considered these two sets of decisions in the first stage of our proposed two-stage stochastic programming models. However, the third set of decisions, i.e., the transportation of casualties to ACFs and hospitals, is made after obtaining more information about uncertain parameters, including (1) *the number of casualties with different injuries in various areas* and (2) *the active treatment capacities of hospitals*, which may differ from their nominal values due to infrastructure damage and operational disruptions. For instance, after the Great East Japan Earthquake in 2011, Fukushima Prefectural Ono Hospital suffered infrastructure damage, initially losing gas, power, and water supplies. This forced the staff to consolidate all patients into a less affected area while continuing care under emergency conditions (Sawano et al., 2023). Similarly, the 2005 Kashmir earthquake led to the complete destruction of 291 healthcare facilities, with an additional 74 partially damaged, severely impacting hospital capacity in the region (Helminen et al., 2006). The 1999 Taiwan earthquake further exemplifies this issue: Christian Hospital experienced a 10% capacity reduction, Veterans Hospital lost 50% of its treatment capacity, and Shiu-Tuan Hospital was completely closed, leaving the largest medical facility in the area unavailable to earthquake victims (Lee and Loh, 1999). Given such real-world instances of hospital infrastructure failures, incorporating uncertainty in treatment capacities is crucial for effective casualty response planning.

We assume that emergency responders will perform a triage and primary survey in the disaster field. This primary survey follows a well-established protocol, represented by the mnemonic A.B.C.D.E., which stands for Airway, Breathing, Circulation, Disability, and Environment. The survey is designed to swiftly identify and address life-threatening conditions while determining the necessary type and extent of medical interventions. For instance, consider a scenario where a victim is alert, able to speak, oriented, and demonstrating movement in all extremities. In such cases, we can confidently conclude that their airway is clear, oxygen is reaching the brain adequately, and there is no significant central neurological injury (Sever et al., 2012). Please note that while we briefly discuss the triage and primary survey here, a detailed examination of this process is outside the scope of this article. Our primary focus is on outlining the care itinerary for high-priority and low-priority casualties within the healthcare network.

We categorize patients into two distinct groups: high-priority and low-priority casualties (Caunhye and Nie, 2018). Each patient group follows a different care itinerary within the healthcare network. High-priority patients are those with breathing problems, significant bleeding, or mental issues that prevent them from following commands. They are prioritized for immediate transportation to hospitals, where they are considered fully treated upon arrival. We assume that hospitals have limited rescue vehicles and treatment capacities but are fully equipped with the necessary renewable and non-renewable resources to treat patients with diverse injuries.

Low-priority patients, however, have a different itinerary. They may need to go through more than one ACF before reaching the status of “fully treated.” Although Caunhye and Nie (2018) assumed that low-priority patients are only treatable at ACFs, we recognize the need for a more adaptable and flexible approach to disaster response. As such, we have considered a refinement to their model, allowing for low-priority patients to initially receive care at ACFs while retaining the option to refer them to either another ACF or a hospital if their condition necessitates further treatment that cannot be provided at the initial ACF. This adjustment addresses the critical issue of hospital over-occupation, particularly in the immediate aftermath of a catastrophic event, which becomes even more pronounced when accounting for casualties who independently seek treatment at hospitals without assistance from emergency responders. Their actions can disrupt response planning and strain valuable resources that could have been allocated to more severe cases (Runge and Buddemeier, 2009). To implement this, first, we transfer low-priority patients to ACFs for the appropriate care and stabilization of their conditions. Their status as fully treated is granted only when the necessary physicians and resources have addressed all of their injuries. This assumption introduces complexity, as ACFs may not always have the essential physicians and medical resources available. In such cases, patients may receive partial treatment at one ACF before being transferred to another ACF or hospital for further care.

Effective management of casualty flows throughout the network is a central aspect of our problem. This includes making decisions about the assignment of patients to specific facilities based on their facility care itinerary. This intricate process involves coordinating the movement of patients to ensure timely access to the necessary care and resources, even when ACFs may lack certain required physicians and medical supplies. Our research focuses on optimizing these facility care itineraries, addressing the diverse nature of patient conditions and resource availability.

In addition to the previously defined patient classification, we may have patients with multiple injuries, further underscoring the need for a flexible approach to care. Therefore, we may need more than one type of physician. Regarding the allocation of physicians to ACFs in the second decision set, we mainly need four types of physicians in order to cover all common injuries after disasters: (i) Orthopedic physicians for fractures, crushes, and other similar injuries, (ii) Wound specialists for lacerations, contusions, and open wounds; (iii) Neurologists for nerve injuries, and (iv) Plastic physicians or dermatologists for burns (Tanaka et al., 1999; Doocy et al., 2013). Physicians require a certain amount of resources to be operational, such as their medical kit and other essential equipment. An injured person may have any combination of the aforementioned injuries. Regarding the allocation of non-renewable resources to ACFs in the second decision set, we consider four classes of them, including drugs, general surgery instruments, orthopedic surgery instruments, and specialized surgery instruments, to treat all mentioned injuries (World Health Organization, 2019). These resources must be supplied by hospitals to ACFs using their limited supply capacities.

The first-stage objective function minimizes the total operational cost of the system, including the establishment cost of ACFs and the transportation cost of non-renewable medical resources from hospitals to ACFs. Moreover, the second-stage objective function minimizes the transportation cost of patients and the penalty corresponding to patients that are not fully recovered due to limited resources in ACFs and hospitals and also the limited number of rescue vehicles.

3. Primary two-stage stochastic mixed-integer programming model (P1)

We propose a two-stage stochastic mixed-integer programming model for the CRP defined in Section 2. The first stage includes decisions on:

- Opening ACFs in potential locations.
- Assigning non-renewable medical resources from hospitals to ACFs.
- Allocating rescue vehicles of different types and physicians with various specialties to ACFs.

Then, after the realization of uncertain demands and hospital treatment capacities, in the second-stage model, we determine the transportation of casualties with varying injuries from different demand points to ACFs and hospitals based on their respective itineraries. High-priority patients receive top priority, ensuring immediate transportation to hospitals, where they are promptly deemed “fully treated” upon arrival. On the other hand, low-priority patients follow a distinct itinerary, which may involve visits to more than one ACF before achieving the “fully treated” status. In the following, we present the first- and second-stage models separately.

3.1. First-stage model

We use the following notation to formulate the first-stage model:

Sets:

$\mathcal{I}$	:	The set of potential locations $i \in \mathcal{I}$ for opening ACFs.
$\mathcal{H}$	:	The set of existing hospitals $h \in \mathcal{H}$.
$\mathcal{M}$	:	The set of rescue vehicle types $m \in \mathcal{M}$.
$\mathcal{J}$	:	The set of possible injuries $j \in \mathcal{J}$.
$\mathcal{J}_i$	:	The subset of injuries that can be treated at ACF $i \in \mathcal{I}$, where $\mathcal{J}_i \subseteq \mathcal{J}$.
$\mathcal{K}$	:	The set of non-renewable resources $k \in \mathcal{K}$.

Parameters:

f_i	:	Fixed cost of opening an ACF at location i .
n_m^{ACF}	:	The available number of rescue vehicles of type m that can be assigned to ACFs.
n_j^{spec}	:	The available number of physicians with specialty j .
c_{ik}^{Inv}	:	The inventory capacity for the non-renewable resource k for an ACF located in i .
c_{hk}^{Supply}	:	The maximum supply capacity of non-renewable resource k from hospital h .
ϕ_k	:	The unit cost associated with the procurement of non-renewable medical resource k for ACFs, covered by the responsible disaster management organization.

First stage variables:

x_i	:	1 if an ACF is opened at location i ; 0 otherwise.
ϑ_{mi}	:	The number of rescue vehicles of type m assigned to ACF at location i .
v_{ij}	:	The number of physicians with specialty j assigned to ACF at location i (Note that ACF at location $i \in \mathcal{I}$ should be able to treat injury type j , i.e., $j \in \mathcal{J}_i$).
r_{ihk}	:	The number of non-renewable resources k assigned to ACF at location i from hospital h .

Using the above notation, we formulate the first-stage model as follows:

$$\min_{\mathbf{x}, \boldsymbol{\vartheta}, \mathbf{v}, \mathbf{r}} \left(\sum_{i \in \mathcal{I}} f_i x_i + \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}} \sum_{k \in \mathcal{K}} \phi_k r_{ihk} + Q(\mathbf{x}, \boldsymbol{\vartheta}, \mathbf{v}, \mathbf{r}) \right) \quad (1)$$

Subject to:

$$\sum_{\substack{i \in \mathcal{I}: \\ j \in \mathcal{J}_i}} v_{ij} \leq n_j^{spec} \quad j \in \mathcal{J} \quad (2)$$

$$\sum_{i \in \mathcal{I}} \vartheta_{mi} \leq n_m^{ACF} \quad m \in \mathcal{M} \quad (3)$$

$$\sum_{h \in \mathcal{H}} r_{ihk} \leq c_{ik}^{Inv} x_i \quad i \in \mathcal{I}, k \in \mathcal{K} \quad (4)$$

$$\sum_{i \in \mathcal{I}} r_{ihk} \leq c_{hk}^{Supply} \quad h \in \mathcal{H}, k \in \mathcal{K} \quad (5)$$

$$\vartheta_{mi} \leq n_m^{ACF} x_i \quad i \in \mathcal{I}, m \in \mathcal{M} \quad (6)$$

$$v_{ij} \leq n_j^{spec} x_i \quad i \in \mathcal{I}, j \in \mathcal{J}_i \quad (7)$$

$$x_i \in \{0, 1\} \quad i \in \mathcal{I} \quad (8)$$

$$\vartheta_{mi}, r_{ihk}, v_{ij} \in \mathbb{Z}^+ \quad i \in \mathcal{I}, h \in \mathcal{H}, m \in \mathcal{M}, k \in \mathcal{K}, j \in \mathcal{J}_i \quad (9)$$

Objective function (1) includes the total fixed cost of opening ACFs, the procurement cost of non-renewable medical resources covered by the responsible disaster management organization, and the second-stage cost $Q(\mathbf{x}, \boldsymbol{\vartheta}, \mathbf{v}, \mathbf{r})$. Here, $Q(\mathbf{x}, \boldsymbol{\vartheta}, \mathbf{v}, \mathbf{r})$ is the expected second-stage cost that is evaluated in the second-stage problem presented in Section 3.2. We have $Q(\mathbf{x}, \boldsymbol{\vartheta}, \mathbf{v}, \mathbf{r}) = E_{s \in \mathcal{S}}[Q(\mathbf{x}, \boldsymbol{\vartheta}, \mathbf{v}, \mathbf{r}, \xi(s))]$ where $E_{s \in \mathcal{S}}[\cdot]$ calculates the expected value over scenario $s \in \mathcal{S}$ and $\xi(s)$ is the vector of uncertain parameters including demands and hospitals' treatment capacity in scenario s . Constraint (2) indicates that the number of physicians with any specialty j assigned to ACFs must be less than or equal to the number of available physicians of the same type. Constraint (3) ensures that the total number of any rescue vehicle of type m assigned to different ACFs cannot exceed the corresponding available number of the same type of rescue vehicles. Constraints (4) and (5) imply the restrictions on the inventory and supply capacity levels of ACFs and hospitals, respectively. Also, according to constraints (6) and (7), we can allocate rescue vehicles and physicians with different specialties only to ACFs that have been established. Constraints (8) and (9) represent integrality constraints for the first-stage variables.

3.2. Second-stage model

To formulate the second-stage model, we use the following notation:

Sets:

- $\mathcal{L}$: The set of demand locations $l \in \mathcal{L}$.
- $\mathcal{P}$: The set of patient classes. This set is the power set of the injury set $\mathcal{J}$ meaning that each member of this set is a subset of the injury set. Each member of $\mathcal{P}$ represents a class of patients with different injuries. For example, $\{1, 3, 4\} \in \mathcal{P}$ represents a class of patients that have the specific injuries 1, 3, and 4.
- S : The set of all possible scenarios $s \in S$.

Parameters:

- c_i^{ACF} : The treatment capacity of ACF i at location i .
- $c_{hs}^{hospital}$: The treatment capacity of hospital h in scenario s .
- c_m^{rescue} : The capacity of the rescue vehicle of type m , measured in terms of transportation time. It represents the total time that a rescue vehicle can spend transporting patients over the considered horizon, multiplied by the number of patients they can carry each time.
- d_{pls}^{low} : The number of low-priority patients of class p realized in demand location l in scenario s .
- d_{pls}^{high} : The number of high-priority patients of class p realized in demand location l in scenario s .
- $n_{mh}^{hospital}$: The available number of rescue vehicles of type m in hospital h .
- t_j : The duration of treatment for a single injury type j .
- τ_j : The available working time of a physician with specialty j in the considered horizon.
- π_{ij} : Estimated travel cost (distance) between point i and point j ($i, j \in \mathcal{J} \cup \mathcal{H} \cup \mathcal{L}$).
- g_{jk} : The amount of non-renewable medical resource k required for the treatment of a single injury type j .
- ρ_{pl}^{low} : Penalty of the unsatisfied low-priority demand of class p at demand location l .
- ρ_{pl}^{high} : Penalty of the unsatisfied high-priority demand of class p at demand location l , where $\rho_{pl}^{low} \ll \rho_{pl}^{high}$.
- $\xi(s)$: The vector of uncertain parameters including $c_{hs}^{hospital}$, d_{pls}^{high} , and d_{pls}^{low} in scenario s .

Second stage variables:

- q_{plims} : The number of patients of class p moved by a rescue vehicle of type m from demand location l to ACF i in scenario s .
- $u_{pp'ihms}$: The number of patients of class p' , who were previously classified as class p , moved by a rescue vehicle of type m from ACF i to hospital h in scenario s .
- ζ_{plhms} : The number of patients of class p moved by a rescue vehicle of type m from demand location l to hospital h in scenario s .
- $\eta_{pp'ii'ims}$: The number of patients of class p' , who were previously classified as class p , moved by a rescue vehicle of type m from ACF i to ACF i' in scenario s .
- μ_{pls}^{low} : Quantity of unsatisfied low-priority demand of class p at demand location l in scenario s .
- μ_{pls}^{high} : Quantity of unsatisfied high-priority demand of class p at demand location l in scenario s .
- σ_{pis} : Patients of class p who are fully treated at ACF i in scenario s (Note that class- p patient should be fully treatable at ACF i).

Based on the given notation, we formulate the second-stage model for each scenario $s \in S$ as follows:

$$Q(x, \theta, v, r, \xi(s)) = \min_{q, u, \eta, \zeta, \mu} \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \left(\sum_{l \in \mathcal{L}} \sum_{i \in \mathcal{J}} \pi_{li} q_{plims} + \sum_{p' \subset p} \sum_{i \in \mathcal{J}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pp'ihms} \right. \\ \left. + \sum_{l \in \mathcal{L}} \sum_{h \in \mathcal{H}} \pi_{lh} \zeta_{plhms} + \sum_{p' \subset p} \sum_{i \in \mathcal{J}} \sum_{i' \in \mathcal{J}} \pi_{ii'} \eta_{pp'ii'ims} \right) \\ + \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(\rho_{pl}^{low} \mu_{pls}^{low} + \rho_{pl}^{high} \mu_{pls}^{high} \right) \quad (10)$$

Subject to:

$$\sum_{i \in \mathcal{J}} \sum_{m \in \mathcal{M}} q_{plims} + \mu_{pls}^{low} = d_{pls}^{low} \quad p \in \mathcal{P}, l \in \mathcal{L} \quad (11)$$

$$\sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \zeta_{plhms} + \mu_{pls}^{high} = d_{pls}^{high} \quad p \in \mathcal{P}, l \in \mathcal{L} \quad (12)$$

$$\sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims} + \sum_{p' \in \mathcal{P}: p \subset p', p' \neq \emptyset} \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pp'ii'ims} = \sigma_{pis} + \sum_{p' \subset p: (p \setminus p') \subseteq \mathcal{J}, p' \neq \emptyset} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} u_{pp'ihms}$$

$$+ \sum_{\substack{p' \subset p: \\ (p \setminus p') \subseteq \mathcal{J}_i, p' \neq \emptyset}} \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pp'ii' ms} \quad i \in \mathcal{J}, p \in \mathcal{P} \quad (13)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} \zeta_{plhms} + \sum_{p \in \mathcal{P}} \sum_{\substack{p' \subset p: \\ p \subset p'}} \sum_{i \in \mathcal{J}} \sum_{m \in \mathcal{M}} u_{pp'ihms} \leq c_{hs}^{hospital} \quad h \in \mathcal{H} \quad (14)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{\substack{p' \subset p: \\ p \subset p'}} \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pp'ii' ms} \leq c_i^{ACF} x_i \quad i \in \mathcal{J} \quad (15)$$

$$t_j \left(\sum_{\substack{p \in \mathcal{P}: \\ j \in p}} \sigma_{pis} + \sum_{\substack{p \in \mathcal{P}: \\ j \in p}} \sum_{\substack{p' \subset p: \\ j \notin p'}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} u_{pp'ihms} + \sum_{\substack{p \in \mathcal{P}: \\ j \in p}} \sum_{\substack{p' \subset p: \\ j \notin p'}} \sum_{\substack{i' \in \mathcal{J}: \\ i' \neq i}} \sum_{m \in \mathcal{M}} \eta_{pp'ii' ms} \right) \leq \tau_j v_{ij} \quad i \in \mathcal{J}, j \in \mathcal{J}_i \quad (16)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \pi_{lh} \zeta_{plhms} \leq c_m^{rescue} n_{mh}^{hospital} \quad m \in \mathcal{M}, h \in \mathcal{H} \quad (17)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \pi_{li} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{\substack{p' \subset p: \\ p' \neq \emptyset}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \pi_{ih} u_{pp'ihms} + \sum_{p \in \mathcal{P}} \sum_{\substack{p' \subset p: \\ p' \neq \emptyset}} \sum_{i' \in \mathcal{J} \setminus \{i\}} \pi_{ii'} \eta_{pp'ii' ms} \leq c_m^{rescue} \vartheta_{mi} \quad m \in \mathcal{M}, i \in \mathcal{J} \quad (18)$$

$$\sum_{p \in \mathcal{P}} \sum_{j \in \{p \cap \mathcal{J}_i\}} \left(g_{jk} \left(\sigma_{pis} + \sum_{\substack{p' \subset p: \\ j \notin p'}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} u_{pp'ihms} + \sum_{\substack{p' \subset p: \\ j \notin p'}} \sum_{\substack{i' \in \mathcal{J}: \\ i' \neq i}} \sum_{m \in \mathcal{M}} \eta_{pp'ii' ms} \right) \right) \leq \sum_{h \in \mathcal{H}} r_{ihk} \quad i \in \mathcal{J}, k \in \mathcal{K} \quad (19)$$

$$q_{plims}, u_{pp'ihms}, \zeta_{plhms}, \eta_{pp'ii' ms}, \mu_{pls}^{low}, \mu_{pls}^{high}, \sigma_{pis} \geq 0, \quad i \in \mathcal{J}, j \in \mathcal{J}_i, m \in \mathcal{M}, \quad h \in \mathcal{H}, k \in \mathcal{K}, p \in \mathcal{P}, l \in \mathcal{L} \quad (20)$$

The objective function (10) represents the second-stage cost in scenario s , including (1) the transportation time of casualties between demand locations, ACFs, and hospitals and (2) the penalties corresponding to not serving some patients due to limited resources. Constraints (11) and (12) allocate low- and high-priority patients to ACFs and hospitals, respectively. In these constraints, unmet patient demands may occur, leading to penalties in the objective function. Constraint (13) indicates that the number of patients of class p admitted to ACF i should match the combined count of patients who have either fully recovered within ACF i or partially recovered at ACF i before being transferred, with a different status $p' \subset p$, to hospitals or other ACFs for additional treatment. The left-hand side of this constraint computes the number of class- p patients entering ACF i from demand points (i.e., $\sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims}$) and other ACFs (i.e., $\sum_{p' \subset p: p \subset p'} \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pp'ii' ms}$). On the other hand, the right-hand side

of this constraint computes the total number of patients that leave ACF i as fully-recovered patients (i.e., σ_{pis}), or are transferred to hospitals (i.e., $\sum_{p' \subset p: (p \setminus p') \subseteq \mathcal{J}_i, p' \neq \emptyset} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} u_{pp'ihms}$) or other ACFs (i.e., $\sum_{p' \subset p: (p \setminus p') \subseteq \mathcal{J}_i, p' \neq \emptyset} \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pp'ii' ms}$) while having a different status $p' \subset p$. The last

two terms on the right-hand side of this constraint ensure that those patients being transferred to hospitals or other ACFs are already treated for at least one of their injuries in ACF i . That is why class- p patients that are not fully treated at ACF i are transferred to hospitals or other ACFs as a new patient class $p' \subset p$. Constraints (14) and (15) ensure that the treatment capacity of hospitals and ACFs is respected. Constraint (16) guarantees that the required treatment time for injury type j at ACF i does not exceed the available time of the corresponding physician at that ACF. It is important to note that we assume each physician treats only one type of injury. Constraints on the rescue vehicle capacity in hospitals and ACFs are formulated by (17) and (18), respectively. On the left-hand side of these constraints, we have computed the total transportation time of patients imposed on the rescue vehicles of ACFs and hospitals, while on the right-hand side, we have the available capacity. Constraint (19) states that the total non-renewable medical resource type k used at ACF i for treating patients must be less than or equal to the amount of this resource supplied by hospitals. Constraint (20) sets the nonnegativity of second-stage decision variables.

Later in our computational results, we will demonstrate a crucial issue with the previously defined second-stage model (P1). Unfortunately, it becomes computationally intractable for realistic-size problems due to its substantial number of variables. Consequently, in the upcoming section, we present a novel and enhanced mathematical model, named model (P2), which significantly outperforms model (P1). In support of this claim, we have established Theorem 1 in online Appendix A, showing the equivalence between these two mathematical models.

4. Improved two-stage stochastic mixed-integer programming model (P2)

In this section, we present an enhanced version of the model (P1) achieved by reducing the dimensionality of the original model, particularly by minimizing the number of decision variables in the second stage. This reduction is accomplished through the manipulation of patient indices within the variables $u_{pp'ihms}$ and $\eta_{pp'ii' ms}$. Sets, parameters, first-stage variables, and a part of second-stage variables remain unchanged. We define the new second-stage variables used in the model (P2) as follows:

- u_{pihms} : The number of patients of class p moved by a rescue vehicle of type m from ACF i to hospital h in scenario s . Here, p refers to the state of the patient while being transferred to the hospital, not its primary state at the ACF.
- $\eta_{pii' ms}$: The number of class- p patients moved by a rescue vehicle of type m from ACF i to ACF i' in scenario s . Here, p refers to the state of the patient while being transferred to ACF i' , not its primary state at ACF i .
- σ_{pjis} : The number of class- p patients at ACF i that are first treated for their injury j in scenario s . These patients could be treated for other injuries in the same ACF, too, but injury j is the first injury type that they are treated for. This variable is the main key for formulating the improved model presented in this section.

The first-stage model (1)–(9) remain unchanged. In the following, we only present the improved second-stage model.

$$Q(x, \vartheta, v, r, \xi(s)) = \min_{q, u, \eta, \zeta, \mu^{low}, \mu^{high}} \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \left(\sum_{l \in \mathcal{L}} \sum_{i \in \mathcal{J}} \pi_{li} q_{plims} + \sum_{i \in \mathcal{J}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pihms} + \sum_{l \in \mathcal{L}} \sum_{h \in \mathcal{H}} \pi_{lh} \zeta_{plhms} \right. \\ \left. + \sum_{i \in \mathcal{J}} \sum_{i' \in \mathcal{J}} \sum_{i' \neq i} \pi_{ii'} \eta_{pii' ms} \right) + \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(\rho_{pl}^{low} \mu_{pls}^{low} + \rho_{pl}^{high} \mu_{pls}^{high} \right) \quad (21)$$

Subject to:

$$\sum_{i \in \mathcal{J}} \sum_{m \in \mathcal{M}} q_{plims} + \mu_{pls}^{low} = d_{pls}^{low} \quad p \in \mathcal{P}, l \in \mathcal{L} \quad (22)$$

$$\sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \zeta_{plhms} + \mu_{pls}^{high} = d_{pls}^{high} \quad p \in \mathcal{P}, l \in \mathcal{L} \quad (23)$$

$$\sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims} + \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pii' ms} + \sum_{j \in \mathcal{J}_i} \sum_{\substack{p' \in \mathcal{P}: \\ p' \setminus \{j\} = p \\ p' \setminus \{j\} \neq \emptyset}} \sigma_{pjis} \\ = \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} u_{pihms} + \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pii' ms} + \sum_{j \in (\mathcal{J}_i \cap p)} \sigma_{pjis} \quad i \in \mathcal{J}, p \in \mathcal{P} \quad (24)$$

$$\sum_{j \in (\mathcal{J}_i \cap p)} \sigma_{pjis} \geq \sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims} + \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pii' ms} \quad i \in \mathcal{J}, p \in \mathcal{P} \quad (25)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} \zeta_{plhms} + \sum_{p \in \mathcal{P}} \sum_{i \in \mathcal{J}} \sum_{m \in \mathcal{M}} u_{pihms} \leq c_{hs}^{hospital} \quad h \in \mathcal{H} \quad (26)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pii' ms} \leq c_i^{ACF} x_i \quad i \in \mathcal{J} \quad (27)$$

$$\sum_{\substack{p \in \mathcal{P}: \\ j \in p}} t_j \sigma_{pjis} \leq \tau_j v_{ij} \quad i \in \mathcal{J}, j \in \mathcal{J}_i \quad (28)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \pi_{lh} \zeta_{plhms} \leq c_m^{rescue} n_{mh}^{hospital} \quad m \in \mathcal{M}, h \in \mathcal{H} \quad (29)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \pi_{li} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pihms} \\ + \sum_{p \in \mathcal{P}} \sum_{i' \in \mathcal{J} \setminus \{i\}} \pi_{ii'} \eta_{pii' ms} \leq c_m^{rescue} \vartheta_{mi} \quad m \in \mathcal{M}, i \in \mathcal{J} \quad (30)$$

$$\sum_{p \in \mathcal{P}} \sum_{j \in \{p \cap \mathcal{J}_i\}} g_{jk} \sigma_{pjis} \leq \sum_{h \in \mathcal{H}} r_{ihk} \quad i \in \mathcal{J}, k \in \mathcal{K} \quad (31)$$

$$q_{plims}, u_{pihms}, \zeta_{plhms}, \eta_{pii' ms}, \mu_{pls}^{low}, \mu_{pls}^{high}, \sigma_{pjis} \geq 0, \quad p \in \mathcal{P}, l \in \mathcal{L}, i \in \mathcal{J}, m \in \mathcal{M}, h \in \mathcal{H}, \\ k \in \mathcal{K}, j \in \mathcal{J}_i, i' \in \mathcal{J}, i' \neq i \quad (32)$$

The objective function (21) minimizes transportation times and penalties corresponding to not serving patients. Constraint (22) ensures that low-priority patients are allocated to ACFs, and constraint (23) assigns high-priority patients to the hospitals. Constraint (24), together with variables σ_{pjis} , are the key components of this model that make it possible to formulate the revised model with significantly fewer variables compared to the primary model (P1). This constraint works as a balance constraint in the improved model. The left-hand side of constraint (24) computes the number of patients of class p in an ACF that are either received from demand locations (i.e., $\sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims}$) and other ACFs (i.e., $\sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pii' ms}$), or are internally generated in the same ACF from treating the first injuries of other patients of classes $p' \in \mathcal{P} |_{p' \setminus \{j\} = p}$, for $j \in \mathcal{J}_i$. The other side of this constraint either transfers these class- p patients to hospitals (i.e., $\sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} u_{pihms}$) or to other ACFs (i.e., $\sum_{i' \in \mathcal{J} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pii' ms}$), or treats them for another injury (i.e., $\sum_{j \in (\mathcal{J}_i \cap p)} \sigma_{pjis}$).

Constraint (25) indicates that all class- p patients sent to ACF i must be treated for at least one injury. The hospital and ACF treatment capacities are respected by constraints (26) and (27), respectively. Physicians of different specialties have limited work time during the planning horizon, which is considered by constraint (28). Constraints (29) and (30) formulate the rescue vehicle capacities for hospitals and ACFs, respectively. Constraint (31) indicates the restriction on the availability of non-renewable medical resources. Constraint (32) declares the nonnegativity of second-stage decision variables.

Figs. 1 and 2 illustrate, through a simple example, the equivalency of the subproblems in models (P1) and (P2), respectively. This scenario involves one hospital and two types of injuries, which leads to $2^2 - 1$ (excluding the empty set) or 3 patient classes—represented as $p = 1$ (e.g., fractures), $p = 2$ (e.g., nerve injuries), and $p = 3$ (both fractures and nerve injuries), with their corresponding binary representations shown in Fig. 2. There is a single type of rescue vehicle for patient transport and one demand location served by two ACFs. ACF #1 is equipped to handle all injury types, while ACF #2 specializes in only injury type #2. Both ACFs are constrained by the limited number of physicians and non-renewable resources available for patient care. As a result, patients may need to be transferred after receiving initial treatment for one injury type to another ACF or hospital for further care, depending on the availability of the required resources and physicians. The yellow cylinders are low-priority patients, and the red circles represent high-priority patients, with p and d indicating their patient class and number of demands, respectively. All patient demands are assumed to be fully satisfied within the depicted scenario.

Theorem 1. Models (P1) and (P2) are equivalent and, when solved, produce the same first-stage optimal solutions with the same optimal objective values.

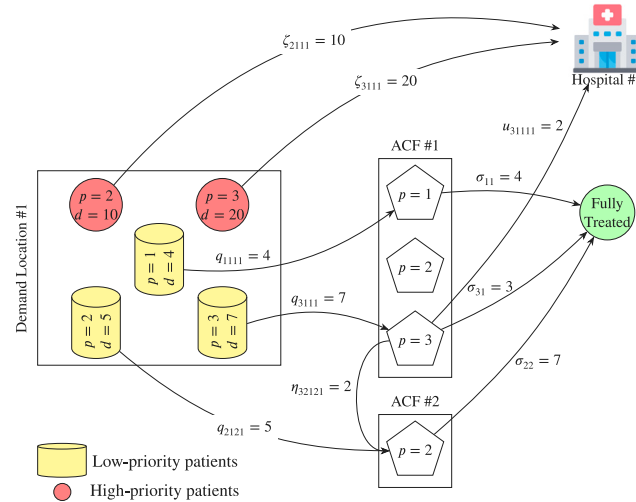


Fig. 1. Network flow of model P1.

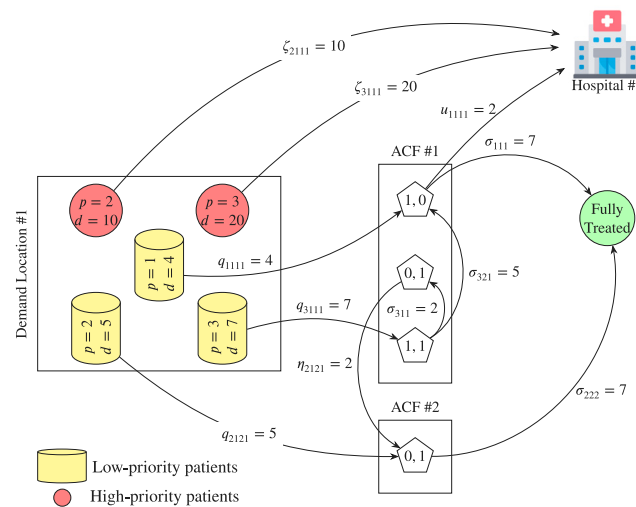


Fig. 2. Network flow of model P2.

Proof. We have provided the proof in Appendix A. The main idea of this proof is that we show that any feasible solution of model (P1) can be projected to a feasible solution of model (P2) with the same first-stage solution and the same objective value, and vice versa. Therefore, both models will result in the same first-stage optimal solution and the same optimal objective value.

5. L-shaped algorithm and enhancements

In this section, we propose an L-shaped algorithm (Van Slyke and Wets, 1969) with various enhancements to efficiently solve the model (P2). The L-shaped method is an iterative approach that alternates between solving the master problem (MP) and subproblems (SPs) for each scenario. After solving the MP, the obtained first-stage solution is fixed in the SPs, which are then solved for all scenarios $s \in S$. Using the dual information from the optimally solved SPs, optimality cuts are generated and added to refine the MP. This process repeats until the maximum acceptable optimality gap is reached or the time limit is met. In the following, we first present the master problem and subproblem of the L-shaped algorithm. We formulate the MP as follows:

$$(\text{MP}) \quad \min_{x, \theta, v, r, \theta} \sum_{i \in \mathcal{I}} f_i x_i + \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}} \sum_{k \in \mathcal{K}} \varrho_k r_{ihk} + \theta \quad (33)$$

Subject to:

$$\begin{aligned}
(2)-(9) \\
\theta &\geq \sum_{s \in S} \phi_s \theta_s \\
\theta_s &\geq \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(d_{pls}^{low} \bar{w}_{spl}^{(1)} + d_{pls}^{high} \bar{w}_{spl}^{(2)} \right) - \sum_{h \in \mathcal{H}} c_{hs}^{hospital} \bar{w}_{sh}^{(5)}
\end{aligned} \tag{34}$$

$$\begin{aligned}
& - \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} c_m^{\text{rescue}} n_{mh}^{\text{hospital}} \bar{w}_{smh}^{(9)} - \sum_{i \in \mathcal{I}} c_i^{\text{ACF}} x_i \bar{w}_{si}^{(6)} - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \tau_j v_{ij} \bar{w}_{sij}^{(7)} \\
& - \sum_{i \in \mathcal{I}} \sum_{m \in \mathcal{M}} c_m^{\text{rescue}} \vartheta_{mi} \bar{w}_{sim}^{(10)} - \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}} \left(\sum_{h \in \mathcal{H}} r_{ihk} \right) \bar{w}_{sik}^{(14)} \quad s \in S \quad (35)
\end{aligned}$$

$$\theta_s \geq 0 \quad s \in S \quad (36)$$

In MP, θ approximates the second-stage cost and ϕ_s represents the probability of scenario s . Optimality cuts (35) are iteratively added after solving SPs as we will discuss later. In constraint (35), $\bar{w}_{spl}^{(1)}$, $\bar{w}_{spl}^{(2)}$, $\bar{w}_{sh}^{(5)}$, $\bar{w}_{si}^{(6)}$, $\bar{w}_{sij}^{(7)}$, $\bar{w}_{smh}^{(9)}$, $\bar{w}_{sim}^{(10)}$, and $\bar{w}_{sik}^{(14)}$ are the optimal dual variables of constraints (22), (23), (26), (27), (28), (29), (30), and (31), respectively. Observe that Benders feasibility cuts are not necessary because the feasibility of the primal subproblem is ensured by constraints (22) and (23). The subproblem (SP_s) is formulated for each scenario $s \in S$ as follows:

$$\begin{aligned}
(SP_s) \quad & \min_{q, u, \eta, \zeta, \mu^{\text{low}}, \mu^{\text{high}}, \sigma} \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \left(\sum_{i \in \mathcal{I}} \sum_{l \in \mathcal{L}} \pi_{li} q_{plims} + \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pihms} + \sum_{l \in \mathcal{L}} \sum_{h \in \mathcal{H}} \pi_{lh} \zeta_{plhms} \right. \\
& \left. + \sum_{i \in \mathcal{I}} \sum_{i' \in \mathcal{I}} \pi_{ii'} \eta_{pii' ms} \right) + \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(\rho_{pl}^{\text{low}} \mu_{pls}^{\text{low}} + \rho_{pl}^{\text{high}} \mu_{pls}^{\text{high}} \right) \quad (37)
\end{aligned}$$

Subject to:

(22)–(26), (29), (32)

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{i' \in \mathcal{I} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pii' ms} \leq c_i^{\text{ACF}} \hat{x}_i \quad i \in \mathcal{I} \quad (38)$$

$$\sum_{p \in \mathcal{P}} \sum_{j \in \mathcal{J}} t_j \sigma_{pjis} \leq \tau_j \hat{v}_{ij} \quad i \in \mathcal{I}, j \in \mathcal{J}_i \quad (39)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \pi_{li} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pihms} + \sum_{p \in \mathcal{P}} \sum_{i' \in \mathcal{I} \setminus \{i\}} \pi_{ii'} \eta_{pii' ms} \leq c_m^{\text{rescue}} \hat{\vartheta}_{mi} \quad m \in \mathcal{M}, i \in \mathcal{I} \quad (40)$$

$$\sum_{p \in \mathcal{P}} \sum_{j \in \{p \cap \mathcal{J}_i\}} g_{jk} \sigma_{pjis} \leq \sum_{h \in \mathcal{H}} \hat{r}_{ihk} \quad i \in \mathcal{I}, k \in \mathcal{K} \quad (41)$$

In the SP, $\hat{x}_i$, $\hat{v}_{ij}$, $\hat{\vartheta}_{mi}$, and $\hat{r}_{ihk}$ are constants that are set to the values of their corresponding variables obtained by the MP. In the remainder of this section, we propose the following eight types of enhancements to accelerate the convergence of our L-shaped algorithm:

- Multiple cuts strategy (Birge and Louveaux, 1988)
- Warm-up strategy (McDaniel and Devine, 1977)
- Lower bounding functional valid inequalities (LBF) (Batun et al., 2011)
- Strong (Pareto-optimal) cuts (PO) (Sherali and Lunday, 2013)
- Benders dual decomposition (BDD) (Rahmaniani et al., 2020)
- Lagrangian dual decomposition (LDD) (Zou et al., 2019)
- Branch-and-Benders-cut scheme (B&BC) (Codato and Fischetti, 2006)
- Heuristic strategies for strengthened cut generation

5.1. Multiple cuts and warm-up strategies

In this section, we discuss two enhancements used to improve our L-shaped algorithm. The first enhancement involves incorporating multiple Benders cuts into the MP at each iteration. There are two strategies to add optimality cuts. In the first strategy, we add the following aggregated cut after solving all SPs:

$$\begin{aligned}
\theta & \geq \sum_{s \in S} \phi_s \left(\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(d_{pls}^{\text{low}} \bar{w}_{spl}^{(1)} + d_{pls}^{\text{high}} \bar{w}_{spl}^{(2)} \right) - \sum_{h \in \mathcal{H}} c_{hs}^{\text{hospital}} \bar{w}_{sh}^{(5)} \right. \\
& - \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} c_m^{\text{rescue}} n_{mh}^{\text{hospital}} \bar{w}_{smh}^{(9)} - \sum_{i \in \mathcal{I}} c_i^{\text{ACF}} x_i \bar{w}_{si}^{(6)} - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \tau_j v_{ij} \bar{w}_{sij}^{(7)} \\
& \left. - \sum_{i \in \mathcal{I}} \sum_{m \in \mathcal{M}} c_m^{\text{rescue}} \vartheta_{mi} \bar{w}_{sim}^{(10)} - \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}} \left(\sum_{h \in \mathcal{H}} r_{ihk} \right) \bar{w}_{sik}^{(14)} \right) \quad (42)
\end{aligned}$$

In constraint (42), $\bar{w}_{spl}^{(1)}$, $\bar{w}_{spl}^{(2)}$, $\bar{w}_{sh}^{(5)}$, $\bar{w}_{si}^{(6)}$, $\bar{w}_{sij}^{(7)}$, $\bar{w}_{smh}^{(9)}$, $\bar{w}_{sim}^{(10)}$, and $\bar{w}_{sik}^{(14)}$ are the optimal dual variables of constraints (22), (23), (26), (27), (28), (29), (30), and (31), respectively. In the second strategy, after solving each subproblem, a single optimality cut is added per scenario:

$$\begin{aligned}
\theta_s & \geq \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(d_{pls}^{\text{low}} \bar{w}_{spl}^{(1)} + d_{pls}^{\text{high}} \bar{w}_{spl}^{(2)} \right) - \sum_{h \in \mathcal{H}} c_{hs}^{\text{hospital}} \bar{w}_{sh}^{(5)} \\
& - \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} c_m^{\text{rescue}} n_{mh}^{\text{hospital}} \bar{w}_{smh}^{(9)} - \sum_{i \in \mathcal{I}} c_i^{\text{ACF}} x_i \bar{w}_{si}^{(6)} - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \tau_j v_{ij} \bar{w}_{sij}^{(7)} \\
& - \sum_{i \in \mathcal{I}} \sum_{m \in \mathcal{M}} c_m^{\text{rescue}} \vartheta_{mi} \bar{w}_{sim}^{(10)} - \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}} \left(\sum_{h \in \mathcal{H}} r_{ihk} \right) \bar{w}_{sik}^{(14)} \quad s \in S \quad (43)
\end{aligned}$$

In this research, we apply the second strategy since, in the literature, it is computationally proven to be more efficient (Batun et al., 2011).

The second enhancement, the warm-up strategy, involves solving the MP as a linear program (LP) for a predefined number of iterations denoted by α^{warm} , or until the number of iterations that the gap is not improved exceeds β^{warm} (McDaniel and Devine, 1977). Then, the integrality constraints are added to the MP and it is solved as a mixed integer linear programming (MILP).

5.2. Lower bounding functional valid inequalities (LBF)

In the following, we present lower bounding functional valid inequalities that improve the lower bound obtained by the master problem. The main idea of LBF valid inequalities is to give the master problem a rough estimation of the second-stage cost based on the information of an average scenario. This approach is in line with the principles of partial Benders decomposition, as shown in the work of Crainic et al. (2021). Based on Jensen's Inequality (Jensen, 1906), we add the following constraints to MP to impose the lower bounding functional:

$$\begin{aligned} \theta \geq & \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \left(\sum_{l \in \mathcal{L}} \sum_{i \in \mathcal{I}} \pi_{li} q_{plim}(\bar{s}) + \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pihm}(\bar{s}) + \sum_{l \in \mathcal{L}} \sum_{h \in \mathcal{H}} \pi_{lh} \zeta_{plhm}(\bar{s}) \right. \\ & \left. + \sum_{i \in \mathcal{I}} \sum_{\substack{i' \in \mathcal{I} \\ i' \neq i}} \pi_{ii'} \eta_{pii'm}(\bar{s}) \right) + \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(\rho_{pl}^{low} \mu_{pl}^{low}(\bar{s}) + \rho_{pl}^{high} \mu_{pl}^{high}(\bar{s}) \right) \end{aligned} \quad (44)$$

$$(22)-(32) \text{ for the average scenario, i.e., } \bar{s}, \text{ rather than scenario } s \quad (45)$$

According to the above relations, the incorporation of LBF valid inequalities requires adding a copy of all second-stage variables and their corresponding constraints for an average scenario to the MP. This utilization of LBF valid inequalities aligns with the principles of partial Benders decomposition, which allows us to generate valid inequalities based on artificial scenarios. Importantly, even if only one such scenario is produced, the best bound that can be obtained is with the average scenario, as demonstrated in Crainic et al. (2021).

5.3. Strong (Pareto-optimal) cuts (PO)

When applying BD, it is common to find multiple optimal solutions when solving the SPs, especially when dealing with network optimization problems such as facility location problems (Magnanti and Wong, 1981). All optimal solutions to the SPs can be used to produce valid cuts for the master problem. However, if the strongest cuts (i.e., Pareto-optimal cuts) are used, the number of iterations and solution time can be significantly decreased. Sherali and Lunday (2013) suggested solving a pre-emptive priority multiple-objective program. According to this method, a perturbed version of the dual subproblem is created to generate Pareto-optimal cuts, represented as model (46). The modification involves incorporating a perturbation term into the dual objective function using a goal-programming weight (ε) and core points of the feasible solution set (i.e., x_i^0 , v_{ij}^0 , g_{mi}^0 , and r_{ihk}^0). When ε takes a small value, for example 10^{-11} in our case, solving the perturbed model (46) yields an optimal solution that corresponds to a Pareto-optimal solution, as stated in Proposition 1.

$$\begin{aligned} \max & \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(d_{pls}^{low} w_{spl}^{(1)} + d_{pls}^{high} w_{spl}^{(2)} \right) - \sum_{h \in \mathcal{H}} c_{hs}^{hospital} w_{sh}^{(5)} - \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} c_m^{rescue} n_{mh}^{hospital} w_{smh}^{(9)} \\ & - \sum_{i \in \mathcal{I}} c_i^{ACF} \hat{x}_i w_{si}^{(6)} - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \tau_j \hat{v}_{ij} w_{sij}^{(7)} - \sum_{i \in \mathcal{I}} \sum_{m \in \mathcal{M}} c_m^{rescue} \hat{g}_{mi} w_{sim}^{(10)} - \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}} \left(\sum_{h \in \mathcal{H}} \hat{r}_{ihk} \right) w_{sik}^{(14)} \\ & + \varepsilon \left(\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(d_{pls}^{low} w_{spl}^{(1)} + d_{pls}^{high} w_{spl}^{(2)} \right) - \sum_{h \in \mathcal{H}} c_{hs}^{hospital} w_{sh}^{(5)} - \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} c_m^{rescue} n_{mh}^{hospital} w_{smh}^{(9)} \right. \\ & \left. - \sum_{i \in \mathcal{I}} c_i^{ACF} x_i^0 w_{si}^{(6)} - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \tau_j v_{ij}^0 w_{sij}^{(7)} - \sum_{i \in \mathcal{I}} \sum_{m \in \mathcal{M}} c_m^{rescue} g_{mi}^0 w_{sim}^{(10)} - \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}} \left(\sum_{h \in \mathcal{H}} r_{ihk}^0 \right) w_{sik}^{(14)} \right) \end{aligned} \quad (46)$$

Subject to:

Constraints for the dual form of the model (21) to (32)

Proposition 1. If we define $opt(\hat{x}, \hat{\delta}, \hat{v}, \hat{r})$ and $sec(\hat{x}, \hat{\delta}, \hat{v}, \hat{r})$ as the optimal and second-optimal values of the dual sub-problem corresponding to the solution of $\hat{x}$, $\hat{\delta}$, $\hat{v}$, and $\hat{r}$, and $opt(x^0, v^0, g^0, r^0)$ as the optimal value of the dual sub-problem associated with x^0 , v^0 , g^0 , and r^0 . Then when the goal-programming weight ε meets the condition $0 < \varepsilon < \frac{sec(\hat{x}, \hat{\delta}, \hat{v}, \hat{r}) - opt(\hat{x}, \hat{\delta}, \hat{v}, \hat{r})}{opt(x^0, v^0, g^0, r^0)}$, an optimal solution to the perturbed model (46) will also be a Pareto-optimal solution.

Proof. We have provided the proof in Appendix B.

5.4. Benders dual approach

We develop the BDD cuts proposed by Rahmaniani et al. (2020) based on Lagrangian duality. In this method, instead of substituting the values of the first-stage solution obtained from MP in the subproblem, we impose the solution to the subproblem as coupling constraints. This process involves creating a new MILP model for the subproblem, where copied variables are introduced to mirror those in the MP with equality constraints (51)–(54). Then, we price out those constraints into the objective function using the associated dual multipliers. By applying this relaxation step, the copied variables no longer equal the first-stage solutions. Consequently, we can enforce integrity constraints on these variables (e.g., constraints (57) and (58)). To develop BDD for our problem, we first replace constraints (38)–(41) in the subproblem with constraints (47)–(55).

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \sum_{m \in \mathcal{M}} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{i' \in \mathcal{I} \setminus \{i\}} \sum_{m \in \mathcal{M}} \eta_{pi'ims} \leq c_i^{ACF} x_i^{Copy} \quad s \in S, i \in \mathcal{I} \quad (47)$$

$$\sum_{\substack{p \in \mathcal{P}: \\ j \in \mathcal{P}}} t_j \sigma_{pjis} \leq \tau_j v_{ij}^{Copy} \quad s \in S, i \in \mathcal{I}, j \in \mathcal{J}_i \quad (48)$$

$$\sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \pi_{li} q_{plims} + \sum_{p \in \mathcal{P}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pihms}$$

$$+ \sum_{p \in \mathcal{P}} \sum_{i' \in \mathcal{I} \setminus \{i\}} \pi_{ii'} \eta_{pii' ms} \leq c_m^{\text{rescue}} \vartheta_{mi}^{\text{Copy}} \quad s \in \mathcal{S}, m \in \mathcal{M}, i \in \mathcal{I} \quad (49)$$

$$\sum_{p \in \mathcal{P}} \sum_{j \in \{p \cap \mathcal{J}_i\}} g_{jk} \sigma_{pjis} \leq \sum_{h \in \mathcal{H}} r_{ihk}^{\text{Copy}} \quad s \in \mathcal{S}, i \in \mathcal{I}, k \in \mathcal{K} \quad (50)$$

$$x_{is}^{\text{Copy}} = \hat{x}_i \quad s \in \mathcal{S}, i \in \mathcal{I} \quad (51)$$

$$v_{ijs}^{\text{Copy}} = \hat{v}_{ij} \quad s \in \mathcal{S}, i \in \mathcal{I}, j \in \mathcal{J}_i \quad (52)$$

$$\vartheta_{mis}^{\text{Copy}} = \hat{\vartheta}_{mi} \quad s \in \mathcal{S}, m \in \mathcal{M}, i \in \mathcal{I} \quad (53)$$

$$r_{ihks}^{\text{Copy}} = \hat{r}_{ihk} \quad s \in \mathcal{S}, i \in \mathcal{I}, h \in \mathcal{H}, k \in \mathcal{K} \quad (54)$$

$$x_{is}^{\text{Copy}}, v_{ijs}^{\text{Copy}}, \vartheta_{mis}^{\text{Copy}}, r_{ihks}^{\text{Copy}} \in \mathbb{R}^+ \quad s \in \mathcal{S}, i \in \mathcal{I}, j \in \mathcal{J}_i, m \in \mathcal{M}, h \in \mathcal{H}, k \in \mathcal{K} \quad (55)$$

Then we let $\lambda_{is}^{\text{Copy}}$, $\lambda_{ijs}^{\text{Copy}}$, $\lambda_{mis}^{\text{Copy}}$, and $\lambda_{ihks}^{\text{Copy}}$ denote the dual multiplier vector corresponding to the constraints (51)–(54), respectively, and form the corresponding Lagrangian relaxation problem as (56)–(58).

$$\begin{aligned} \max_{\lambda^{\text{Copy}}, \lambda^{\text{Copy}}, \lambda^{\text{Copy}}, \lambda^{\text{Copy}} \in \mathbb{R}^n} \min \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \left(\sum_{l \in \mathcal{L}} \sum_{i \in \mathcal{I}} \pi_{li} q_{plims} + \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}} \pi_{ih} u_{pihms} + \sum_{l \in \mathcal{L}} \sum_{h \in \mathcal{H}} \pi_{lh} \bar{z}_{plhms} \right. \\ \left. + \sum_{i \in \mathcal{I}} \sum_{i' \in \mathcal{I}} \pi_{ii'} \eta_{pii' ms} \right) + \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(\rho_{pl}^{\text{low}} \mu_{pls}^{\text{low}} + \rho_{pl}^{\text{high}} \mu_{pls}^{\text{high}} \right) - \left(\lambda_{is}^{\text{Copy}} (x_{is}^{\text{Copy}} - \hat{x}_i) \right. \\ \left. + \lambda_{ijs}^{\text{Copy}} (v_{ijs}^{\text{Copy}} - \hat{v}_{ij}) + \lambda_{mis}^{\text{Copy}} (\vartheta_{mis}^{\text{Copy}} - \hat{\vartheta}_{mi}) + \lambda_{ihks}^{\text{Copy}} (r_{ihks}^{\text{Copy}} - \hat{r}_{ihk}) \right) \end{aligned} \quad (56)$$

Subject to:

$$(2)-(9), (22)-(26), (29), (32), (47)-(50)$$

$$x_{is}^{\text{Copy}} \in \{0, 1\} \quad i \in \mathcal{I} \quad (57)$$

$$\vartheta_{mis}^{\text{Copy}}, v_{ijs}^{\text{Copy}}, r_{ihks}^{\text{Copy}} \in \mathbb{Z}^+ \quad i \in \mathcal{I}, m \in \mathcal{M}, h \in \mathcal{H}, k \in \mathcal{K}, j \in \mathcal{J}_i \quad (58)$$

Note that, in the model (56)–(58), we apply the integrality requirements on x_{is}^{Copy} , v_{ijs}^{Copy} , $\vartheta_{mis}^{\text{Copy}}$, and r_{ihks}^{Copy} . Then, for any values of $\lambda_{is}^{\text{Copy}}$, $\lambda_{ijs}^{\text{Copy}}$, $\lambda_{mis}^{\text{Copy}}$, and $\lambda_{ihks}^{\text{Copy}}$, after solving the inner minimization of the Lagrangian relaxation model (56)–(58), we can add the strengthened cut (59) to the MP. Notably, in BDD, the strengthened cut serves as a valid lower bound, eliminating the need to optimize the Lagrangian multipliers.

$$\begin{aligned} \bar{\vartheta}^s \geq \sum_{p \in \mathcal{P}} \sum_{m \in \mathcal{M}} \left(\sum_{l \in \mathcal{L}} \sum_{i \in \mathcal{I}} \pi_{li} \bar{q}_{plims} + \sum_{i \in \mathcal{I}} \sum_{h \in \mathcal{H}} \pi_{ih} \bar{u}_{pihms} + \sum_{l \in \mathcal{L}} \sum_{h \in \mathcal{H}} \pi_{lh} \bar{z}_{plhms} + \sum_{i \in \mathcal{I}} \sum_{i' \in \mathcal{I}} \pi_{ii'} \bar{\eta}_{pii' ms} \right) \\ + \sum_{p \in \mathcal{P}} \sum_{l \in \mathcal{L}} \left(\rho_{pl}^{\text{low}} \bar{\mu}_{pls}^{\text{low}} + \rho_{pl}^{\text{high}} \bar{\mu}_{pls}^{\text{high}} \right) - \left(\lambda_{is}^{\text{Copy}} (x_i - \bar{x}_i^{\text{LR}}) + \lambda_{ijs}^{\text{Copy}} (v_{ij} - \bar{v}_{ij}^{\text{LR}}) \right. \\ \left. + \lambda_{mis}^{\text{Copy}} (\vartheta_{mi} - \bar{\vartheta}_{mi}^{\text{LR}}) + \lambda_{ihk}^{\text{Copy}} (r_{ihk} - \bar{r}_{ihk}^{\text{LR}}) \right) \end{aligned} \quad (59)$$

In the strengthened cut (59), $\bar{x}_i^{\text{LR}}$, $\bar{v}_{ij}^{\text{LR}}$, $\bar{\vartheta}_{mi}^{\text{LR}}$, and $\bar{r}_{ihk}^{\text{LR}}$ are parameters that have already been determined in the Lagrangian relaxation problem. [Rahmaniani et al. \(2020\)](#) showed that when the optimal solution for the MP is an integer solution, the optimality cut (59) is at most as strong as the general optimality cut (43). However, when dealing with fractional solutions in the MP, the strengthened cut (59) can outperform the generalized Benders cut (43).

We also investigated the utilization of the LDD method. In this approach, in dealing with (56)–(58), we first optimize the Lagrangian multiplier values, i.e., $\lambda_{is}^{\text{Copy}}$, $\lambda_{mi}^{\text{Copy}}$, $\lambda_{ij}^{\text{Copy}}$, $\lambda_{ihk}^{\text{Copy}}$, through the heuristic model (60)–(61) and then generate the cut (59). We refer to the obtained cuts as LDD cuts.

$$\max_{\eta^{\text{Heuristic}} \in \mathbb{R}^1, \lambda \in \mathbb{R}^n} \left(\eta^{\text{Heuristic}} - \frac{\delta}{2} \|\lambda^{(t-1)} - \lambda\|^2 \right) \quad (60)$$

Subject to:

$$\eta^{\text{Heuristic}} \leq SP_{obj^V} + (\bar{y} - \bar{z}^V) \lambda \quad V = 1, 2, \dots, t-1 \quad (61)$$

The objective function (60) aims to increase the amount of cut lifting minus the distance value. The distance value allows the algorithm to find a new vector λ^t close to the vector obtained from the previous iteration $\lambda^{(t-1)}$. In this model, $\bar{y}$ is the vector of fractional values corresponding to the first-stage variable fixed in the MP. The parameter δ is selected as the step size. In (61), SP_{obj^V} denotes the value of the objective function (56) without the Lagrangian penalties obtained through solving the Lagrangian relaxation model in the V^{th} iteration. Also, $\bar{z}^V$ represents the value vector of the corresponding first-stage variables. After solving the heuristic model (60)–(61), we have the new Lagrangian multiplier values, build the inner minimization problem (56)–(58), and obtain the relaxed first- and second-stage variables. We then add a new constraint to (61) and solve the revised model (60)–(61) again. This process continues until either the new multiplier values show less improvement than the prior iteration or the maximum iteration limit is reached.

5.5. Branch-and-benders-cut scheme

In the classical BD, a single MILP problem is solved to optimally at each iteration. This requires constructing a new branch-and-bound tree each time, leading to potential inefficiencies as previously eliminated candidate solutions are revisited. In this article, whenever we mention “*L-shaped*,” we are actually referring to the classical BD, which is a pure cutting-plane method applied to our stochastic mathematical model. An alternative approach involves constructing a single search tree and generating constraints for both integer and fractional solutions encountered within the tree. This alternative approach leads to the same optimal solution and is commonly known as the branch-and-Benders-cut scheme. In this article, we will refer to this alternative method as the “*Branch-and-Benders-Cut (B&BC)*” algorithm, in which we add cuts for fractional solutions only at the root node, and then only when new incumbent solutions are found. These cuts and valid inequalities are incorporated into the problem using the Callback procedures available in the IBM CPLEX Optimization Solver’s Concert Technology library. We will apply both the “*L-shaped*” and “*B&BC*” algorithms and compare the results in Section 6.3 and online Appendix F.

5.6. Heuristic strategies for strengthened cut generation

In this section, we introduce two enhanced cut generation strategies aimed at reducing computational effort, particularly when optimizing the Lagrangian dual problem for BDD and LDD cuts, as detailed in Section 5.4. These acceleration techniques, including the warm-up, BDD, and LDD, are exclusively applied at the root node of the B&C tree (where at most half of the time limit is dedicated to solving the LP at the root node). However, Modifications are still needed to reduce the computational burden of generating strengthened cuts, as solving a large number of mixed-integer programming (MIP) subproblems at each iteration is time-consuming. This is because the method introduced by Rahmaniani et al. (2020) was meant for a single-scenario problem, while we aim to solve a stochastic problem with at least 100 scenarios. It is important to note that, apart from the root node, we use only generalized Benders cuts (43) at all other nodes in the tree.

The first strategy, which we refer to as BBC_I , is inspired by the three-phase approach proposed by Rahmaniani et al. (2020). In this strategy, we first quickly obtain valid cuts by solving the LP relaxation of the MP. We do this because, during the early stages of the Benders algorithm, the solutions of the MP tend to have low quality. Subsequently, we follow an iterative approach by solving the LP relaxation of the MP and SPs while incorporating generalized Benders cuts in each iteration. We continue this process for a specific number of iterations ($\alpha^{warm} = 15$) or until the number of iterations with unimproved gaps exceeds a predefined threshold ($\beta^{warm} = 3$). Then, we create strengthened Benders cuts, as explained in Section 5.4 considering the modified version. In the modified version for BDD and LDD, when we encounter a fractional solution in the MP at the root node, we do not solve all SPs to produce strengthened cuts for every scenario. Instead, we tackle only a subset of them in each iteration. For example, in the first iteration, we solve the first $\alpha^{cluster} = 10\%$ of scenarios; in the second iteration, we solve the next $\alpha^{cluster} = 10\%$, and so on. This process is repeated for a limited number of iterations ($\alpha^{BDD} = 10$) or when the B&BC gap shows no improvement over three consecutive iterations ($\beta^{BDD} = 3$).

In our second strategy, known as BBC_{II} , we employ a unique approach by applying the warm-up process, BDD, and LDD consecutively but on specific scenario clusters. This means that when we encounter a fractional solution in the master problem at the root node, we handle one cluster at a time. For instance, in the first iteration, we focus on the initial $\alpha^{cluster} = 10\%$ of scenarios, utilizing the warm-up process along with the corresponding generalized Benders cuts. In the subsequent iteration, we move to the next $\alpha^{cluster} = 10\%$ of scenarios and apply strengthened Benders cuts tailored to this cluster, and so on. This process of addressing each cluster one at a time is repeated until we reach a set number of iterations ($\alpha^{BBC_{II}} = 15$) or experience a failure to reduce the gap for three consecutive iterations ($\beta^{BBC_{II}} = 3$). So, in essence, we employ the warm-up, BDD, and LDD techniques individually for each cluster rather than consecutively, making it a more step-by-step or “layered” approach. In BBC_{II} , the heuristic model (60)–(61) is terminated either if we reach the maximum number of iterations, $\alpha^{Heuristic} = 10$, or if we fail to lift the cut by at least $\gamma^{Heuristic} = 10^{-1}$. We have also selected the following values for the remaining parameters in this model: $\delta = 10^5$ and $\nabla = 10^2$. We choose a large number for δ since we aim to remain close to what we obtained as Lagrangian multiplier values after dualizing of constraints (51)–(54).

We have also developed two improvement methods, namely partially relaxed subproblems and ϵ -optimal cuts, for both strategies. In the former method, when we are applying the BDD or LDD, the integrality constraints are only imposed on a subset of first-stage variables, i.e., x_i and θ_{mi} . In that case, we produce a weaker optimality cut compared to the Lagrangian cut due to relaxed integrality constraints on variables v_{ij} and r_{ihk} . However, it still provides a stronger cut than a general Benders cut. In the latter method, we stop solving the MIP subproblems in BDD and LDD once we reach the ϵ -optimal solutions. These solutions have objective values no more than $\epsilon = 10^{-2}$ units higher than the lower bound. The overall implementation of the BBC_I algorithm is illustrated as a flowchart in online Appendix C.

6. Computational results

We carried out the computational tests on a machine with two 2.40 GHz Intel Xeon Gold 6148 processors using C++ and IBM ILOG CPLEX Optimization Studio V22.1 software. One core was used to execute each test instance. In Section 6.1, we describe the process by which the set of CRP instances with uncertain demand and hospital treatment capacity was obtained. In Section 6.2, we compare the performance of the two proposed mathematical models, and then we assess the computational efficiency of the different enhancements used in our algorithm in Sections 6.3. The effectiveness of each accelerator is also evaluated in Section 6.4. Finally, in Section 6.5, we analyze the impact of casualty shortage penalty parameters on model behavior.

6.1. Instance generation

In this section, we explain the details of generating CRP instances with uncertain demand and hospital treatment capacity. For generating a major part of our instances, we used the data provided by Oksuz and Satoglu (2020) and Sun et al. (2021). However, since we consider some details in CRPs for the very first time, we needed to generate the newly defined parameters as detailed later in this section. *Parameter Settings*: We set the number of ACFs and demand locations to $\{10, 15, 20, 25, 30\}$ and the number of hospitals to $\{10\}$. We assume there are three types of rescue vehicles with capacities of 1, 2, and 3 patients per trip (Rambha et al., 2021).

Table 2

Parameters used for instance generation.

Parameters	Value	Reference
Probability of occurrence of scenario s	$\phi^s = \frac{1}{ S }$	
The treatment capacity of ACF i	$c_i^{ACF} = U[400, 1200]$	Oksuz and Satoglu (2020)
Fixed cost of opening an ACF at point i	$f_i = c_i^{ACF} \times 1.3 \times 25$	Oksuz and Satoglu (2020)
The treatment capacity of hospital h in scenario s	$c_{hs}^{hospital} = U[1000, 2000]$	Caunhye et al. (2015)
The inventory capacity for non-renewable resource k at ACF i	$c_{ik}^{Inv} = c_i^{ACF} \times U[7, 9]$	Sun et al. (2021)
The maximum supply capacity of non-renewable resource k from hospital h	$c_{hk}^{Supply} = c_{hs}^{hospital} \times U[14, 16]$	Sun et al. (2021)
Penalty cost of the unsatisfied high-priority demand of class p at location l	$\rho_{pl}^{high} = \$150,000$	Li et al. (2020)
Penalty cost of the unsatisfied low-priority demand of class p at location l	$\rho_{pl}^{low} = \frac{\rho_{pl}^{high}}{2}$	

We uniformly generated the demand locations, ACFs, and hospitals in a square area with a side length of 50 km. For each demand location, we generated the number of patients by a uniform distribution of [50, 200]. We use the uniform distribution to generate demand and other uncertain parameters because our model is intended to be general and applicable to various disaster scenarios rather than being tailored to a specific event. Different disasters may lead to different demand distributions, but by using a uniform distribution over a realistic range of values, we ensure that our scenario generation captures a broad spectrum of possibilities. This approach provides a diverse set of instances, allowing us to evaluate the performance of our algorithm across a wide range of possible conditions. Additionally, the uniform distribution is well-suited when there is limited information about the actual demand distribution (Jr and Taskin, 2008). In casualty response planning, demand estimation immediately after a disaster is often uncertain, making the uniform distribution a reasonable choice. Furthermore, within a given range, the uniform distribution is the maximum entropy distribution, meaning that it makes the fewest assumptions about the likelihood of specific values and treats all possibilities equally (Wu et al., 2024). While specific events may require different distributions, our model remains flexible and can be adjusted accordingly for particular disaster cases.

Regarding the injuries of patients, we considered four common categories of disaster-related injuries, including fractures/crush (FC), laceration/contusions/open wounds (LCO), peripheral nerve injury (PNI), and burns (BU). The occurrence probabilities of these injuries are 63%, 33%, 2%, and 2% of injuries in disasters, respectively (Tanaka et al., 1999; Doocy et al., 2013). To determine which patient suffers from which injuries, we first determine the number of injuries for each patient utilizing an integer uniform distribution [1, 4]. Then, we generate the patients' injuries using the above occurrence probabilities. This results in $2^4 - 1$ categories of patients, including all possible injury subsets except the empty set. Moreover, Sun et al. (2021) supposed that 40% to 45% of patients are high-priority. In our study, we assume that patients with 1, 2, 3, and 4 injuries are categorized as high-priority patients, with a uniform distribution range of 20 – 30, 35 – 40, 45 – 50, and 55 – 60, respectively.

For the duration of treatments, we used the means and standard deviations for orthopedics, general surgeries, neurosurgery, and plastic surgeries provided by Costa (2017). The non-renewable resources are also classified into four general groups of drug kits and three other surgical kits for general, orthopedic, and specialized surgeries (World Health Organization, 2019). The cost of these kits is \$7.5 for drugs and \$13.5 for other surgical kits (Sun et al., 2021). All patients require the first group of non-renewable resources, while using the other kits depends on the type of injury. We provide the rest of the parameters in Table 2.

6.2. Performance of mathematical models

As discussed in Section 3.2, we provided an improved model (P2) that is an enhanced version of the basic mathematical model (P1) for the class of CRPs with multiple injuries. In Table 3, we have compared the performance of these two mathematical models. In the instances of Table 3, we have set the number of scenarios and hospitals to 10. Each row of this table presents the average value for 5 different instances. The first two columns of Table 3 show instances' information, including the number of ACFs and the number of demand points. Under “# Vars.” and “# Const.”, we have provided the number of variables and constraints of each model. Under the “Time (s)” column, we report the time required to solve the problem optimally using CPLEX directly. We have reported the three recent columns for both models (P1) and (P2). Finally, the last column reports the improvement in the solution time resulting from using model (P2) instead of model (P1), i.e., $\Delta^{\text{time}} = 100 \times \left(\frac{P1's \text{ time} - P2's \text{ time}}{P1's \text{ time}} \right)$. Table 3 shows that model (P2) significantly outperforms model (P1) in terms of solution time and is, on average, 42.56% faster. This saving in solution time is mainly a result of the reduction in the number of variables by approximately 88% in the model (P2) compared to (P1). Since model (P2) shows significantly stronger performance, we present all numerical results in the following sections based on this model.

6.3. Performance of algorithms

The main goal of this subsection is to assess the performance of accelerators and compare the two strengthened cut generation strategies explained in Section 5.6. Table 4 presents the computational results for the B&BC algorithm and PHA, which serves as our benchmark (detailed information on PHA is provided in online Appendix D). Specifically, BBC_0 is the basic implementation of the B&BC scheme, excluding any additional acceleration techniques as outlined in Section 5. In contrast, BBC_I and BBC_{II} are advanced versions of the algorithm. These incorporate all enhancement techniques discussed from Sections 5.1 through 5.5, in addition to two distinct strengthened cut generation strategies detailed in Section 5.6.

Each row of the Table 4 represents the average results across five instances. We have considered a time limit of 24 h for solving instances with these four algorithms. In this table, the first two columns provide the number of potential locations for the ACFs and the number of demand locations, respectively. “Nodes No.” and “Time (s)” refer to the number of explored nodes in the branch-and-bound tree and the total solution time, respectively. Column “Gap (%)” computes the gap between the lower- and the upper-bound, which are presented under columns “LB” and “UB” respectively. Additionally, for PHA, we include “No. Iter.”, which represents the total number of iterations. Column “Obj.” provides the final objective function value obtained using PHA. Lastly, “ $\Delta^{Obj.}$ ” measures the relative objective improvement achieved by BBC_I over PHA, calculated

Table 3

Comparison of mathematical models P1 and P2.

Problem size		P1 model			P2 model			Δ^{time} (%)
# ACF	# De.Loc.	# Vars.	# Const.	Time (s)	# Vars.	# Const.	Time (s)	
20	10	4,056,924	8671	34,578	403,444	11,671	18,303	47.79
	15	4125,926	10,195	19,050	472,622	13,195	9320	48.61
	20	4,194,924	11,669	24,422	541,428	14,669	16,326	33.33
	25	4,263,928	13,195	36,833	610,622	16,195	18,749	41.14
	30	4,332,922	14,682	21,363	679,525	17,682	10,924	44.01
25	10	5,902,912	10,074	42,414	549,268	13,824	23,610	42.89
	15	5,983,156	11,479	41,463	628,821	15,229	24,902	37.13
	20	6,063,406	12,973	35,645	709,023	16,723	19,174	43.57
	25	6,143,656	14,486	52,620	789,370	18,236	29,359	44.92
	30	6,223,908	15,997	54,769	869,701	19,747	29,810	44.93
30	10	8,086,392	11,318	54,778	716,425	15,818	26,652	51.38
	15	8,177,885	12,752	67,389	807,439	17,252	39,199	41.59
	20	8,269,391	14,327	60,052	899,490	18,827	37,193	40.38
	25	8,360,882	15,754	62,961	990,455	20,254	38,681	38.00
	30	8,452,387	17,292	63,979	1,082,231	21,792	39,074	38.80
Average		6,175,907	12,991	44,821	716,658	16,741	25,418	42.56

Note. To save space, the results for #ACF = 10 and #ACF = 15 are not presented in this table.

Table 4

Computational results of the BBC Algorithm for the CRP problem with stochastic demand and hospital treatment capacity.

Instance		BBC_0			BBC_{II}		BBC_I										PHA					
		Nodes	No.	Time (s)	Gap (%)	Time (s)	Gap (%)	Nodes	No.	Time (s)	LB	UB	Gap (%)	EC (%)	PC (%)	TC (%)	SC (%)	VSS (%)	No. Iter.	Time (s)	Obj.	$\Delta^{Obj.}$ (%)
10	10	20,352	8742	0.0	2398	0.0	2673	2082	71,294	71,294	0.0	55.8	21.6	22.5	0.0	95.98	317	86,400	17,836,640	99.25		
	15	57,222	29,572	0.0	3102	0.0	4517	3206	84,936	84,936	0.0	47.1	26.1	26.8	0.0	95.88	205	86,400	10,205,713	95.34		
	20	77,248	65,904	12.7	4740	0.0	6703	6453	112,396	112,396	0.0	44.5	26.8	28.7	0.0	95.25	151	86,400	12,967,317	95.6		
	25	64,102	86,400	37.4	4381	0.0	2512	3674	134,560	134,560	0.0	43.5	25.9	30.7	0.0	94.97	119	86,400	2,269,492	83.23		
	30	53,991	86,400	72.6	5911	0.0	6015	9149	152,864	152,864	0.0	44.0	27.1	28.9	0.0	94.83	91	86,400	3,950,736	86.07		
15	10	44,817	52,929	32.50	3620	0.0	4670	3863	70,144	70,144	0.0	54.2	23.4	22.5	0.0	95.84	128	86,400	16,349,430	99.43		
	15	49,260	74,733	49.90	7013	0.0	4887	3492	86,515	86,515	0.0	45.3	24.8	28.2	1.7	95.70	100	86,400	11,947,294	98.74		
	20	51,057	86,400	105.4	5800	0.0	7152	8848	106,614	106,614	0.0	45.2	26.4	28.4	0.0	95.69	73	86,400	1,635,050	89.71		
	25	40,426	86,400	174.7	11,442	0.0	8099	11,198	129,213	129,213	0.0	44.7	27.1	28.2	0.0	95.43	55	86,400	2,861,579	86.06		
	30	36,343	86,400	176.4	13,265	0.0	6218	11,585	148,799	148,799	0.0	45.9	27.9	26.3	0.0	95.25	46	86,400	826,801	81.19		
20	10	44,755	82,091	82.1	8209	0.0	8027	8210	89,341	89,341	0.0	52.5	24.9	22.5	0.0	96.46	64	86,400	19,528,625	99.50		
	15	46,615	86,400	83.5	6809	0.0	3719	6333	81,913	81,913	0.0	44.4	26.4	29.2	0.0	96.24	52	86,400	17,980,166	99.03		
	20	33,866	86,400	193.7	14,799	0.0	3447	11,397	135,098	135,098	0.0	45.1	27.7	27.2	0.0	96.40	43	86,400	1,806,150	87.73		
	25	37,311	86,400	219.5	17,773	0.0	8083	16,728	129,052	129,052	0.0	45.1	27.3	27.6	0.0	95.28	30	86,400	2,391,572	86.32		
	30	28,487	86,400	493	39,170	0.0	9713	19,861	189,248	189,248	0.0	44.1	27.9	28.0	0.0	95.58	24	86,400	1,127,985	81.65		
25	10	38,567	86,076	174.4	11,123	0.0	4639	9288	62,969	62,969	0.0	49.2	24.6	26.2	0.0	96.47	36	86,400	24,211,720	99.74		
	15	31,126	86,400	224.0	10,572	0.0	4412	12,310	81,884	81,884	0.0	44.0	26.3	29.7	0.0	96.06	26	86,400	13,590,664	94.87		
	20	30,653	86,400	190.9	25,843	0.0	2889	20,392	103,774	103,774	0.0	43.8	27.0	29.3	0.0	95.70	19	86,400	7,763,895	93.27		
	25	28,500	86,400	827.9	51,736	0.6	11,691	34,941	127,287	127,287	0.0	43.8	27.7	28.5	0.0	95.19	16	86,400	10,000,720	86.79		
	30	22,462	86,400	255.0	52,985	2.9	10,058	51,805	144,366	146,494	1.5	44.4	28.3	27.2	0.1	95.27	13	86,400	3,344,404	84.29		
30	10	24,411	86,400	410.6	19,230	0.0	2924	24,869	65,726	65,726	0.0	52.7	22.9	24.4	0.0	96.22	21	86,400	14,594,000	99.42		
	15	28,085	86,400	208.9	32,332	0.0	7054	34,350	82,406	82,406	0.0	46.7	26.5	26.8	0.0	96.13	14	86,400	6,311,763	87.81		
	20	25,575	86,400	256.5	40,186	0.0	8453	46,085	101,943	102,028	0.1	45.4	28.1	26.5	0.0	95.81	11	86,400	16,734,792	94.20		
	25	21,031	86,400	1897	77,125	4.0	9719	57,124	124,152	125,422	1.1	43.5	27.7	28.8	0.0	95.37	9	86,400	5,555,426	82.77		
	30	17,531	86,400	7899	83,403	2.4	7879	62,579	157,632	159,152	1.0	45.0	28.3	26.8	0.0	95.36	8	86,400	711,119	77.01		
Average		38,152	78,210	563.1	22,119	0.4	6246	19,193	110,965	111,165	0.1	46.4	26.4	27.2	0.07	95.69	67	86,400	9,060,122	90.76		

Note. To save space, details on LB and UB for the BBC_0 and BBC_{II} approaches are not presented in this table.

as $100 \times \left(\frac{Obj_{PHA} - UB_{BBC_I}}{Obj_{PHA}} \right)$. This metric highlights the percentage reduction in objective function value when using our proposed approach compared to the benchmark.

Since the BBC_I has resulted in the best performance, we reported more information for this algorithm. “EC (%)”, “PC (%)”, “TC (%)”, and “SC (%)” give the contribution percentage of establishment cost, procurement cost, transportation cost, and penalty cost in the total cost for the solutions by solving the stochastic model using the BBC_I algorithm. “VSS (%)” also computes the value of the stochastic solution in percentage. This value is computed as follows: $VSS = 100 \times \left(\frac{EEV - UB}{EEV} \right)$, where EEV is the expected objective value of the expected value solution computed using the scenario set. To obtain EEV, we first solve a deterministic problem using the expected value for the stochastic parameters, also known as the “mean-value” problem. Then, we save the optimal variables obtained in the first stage, fix them in the stochastic model, and proceed to solve the second stage for the different scenarios.

According to Table 4, BBC_0 , BBC_{II} , and BBC_I algorithms have resulted in an average optimality gap of 563.1%, 0.4%, and 0.1%, respectively. These results show that the proposed acceleration techniques significantly contribute to improving the performance of BBC_I and BBC_{II} compared to the BBC_0 algorithm. For the largest instances, the average optimality gap of the BBC_0 algorithm has skyrocketed to 7900%, while we have obtained 2.4% and 1.0% gaps for BBC_{II} and BBC_I , highlighting the accelerators’ effectiveness. Both BBC_I and BBC_{II} algorithms performed well and found optimal solutions in 21 instance settings. In other instances, BBC_I performed better than BBC_{II} in 3 cases, whereas BBC_{II} outperformed BBC_I on a single instance. The solution time analysis reveals that BBC_I achieved the optimal solutions faster when compared to BBC_{II} in 57% of the cases. Moreover, both BBC_I and BBC_{II} algorithms performed better than the BBC_0 approach in terms of solution time across all instances.

Comparing our best algorithm, BBC_I , with the benchmark PHA, we observe a significant improvement in both computational efficiency and solution quality. PHA failed to converge within the 24-h time limit for even the smallest instances, with an average solution time of 86,400 s, whereas BBC_I required only 19,193 s on average. This highlights the efficiency of our proposed approach in reaching high-quality solutions within a practical computational timeframe. Furthermore, BBC_I achieved an average objective function value of 111,165, whereas PHA resulted in an objective value of 9,060,122. This corresponds to a 90.76% improvement in the objective function when transitioning from PHA to BBC_I , demonstrating the superiority of our algorithm in solving the CRP problem effectively.

Table 5
Reverse marginal impact.

Problem size		$BBC_{I \setminus \{LBF\}}$		$BBC_{I \setminus \{PO\}}$		$BBC_{I \setminus \{Warm-up\}}$		$BBC_{I \setminus \{BDD\}}$		$BBC_{I \setminus \{LDD\}}$	
# ACF	# De.Loc.	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)	Time (s)	Gap (%)
10	10	4855	0.0	1540	0.0	2488	0.0	1844	0.00	5319	0.00
	15	10,384	0.0	2764	0.0	2762	0.0	2605	0.00	6300	0.00
	20	13,379	0.0	4521	0.0	4509	0.0	3530	0.00	9270	0.00
	25	20,115	0.0	5554	0.0	5872	0.0	3381	0.00	15,466	0.00
	30	42,556	0.0	7104	0.0	5601	0.0	4649	0.00	14,015	0.00
15	10	8614	0.0	4584	0.0	3140	0.0	4427	0.00	13,774	0.00
	15	29,093	0.0	4931	0.0	6798	0.0	6978	0.00	15,937	0.00
	20	51,963	5.1	7919	0.0	4738	0.0	5831	0.00	40,206	0.00
	25	80,137	15.9	7013	0.0	9182	0.0	8818	0.00	44,029	0.00
	30	78,381	24.0	13,473	0.0	11,200	0.0	14,692	0.00	44,741	0.00
20	10	34,132	0.0	18,346	0.0	11,284	0.0	11,795	0.00	43,523	0.00
	15	31,421	0.0	6654	0.0	5022	0.0	4444	0.00	36,722	0.00
	20	86,400	25.2	13,144	0.0	12,985	0.0	11,877	0.00	54,277	0.00
	25	86,400	52.2	28,912	0.3	25,119	0.0	15,468	0.00	52,442	0.00
	30	86,400	56.0	38,802	1.8	35,559	0.0	20,069	0.00	56,968	0.76
25	10	40,512	18.5	11,671	0.0	8481	0.0	6725	0.00	49,882	0.00
	15	67,763	10.8	11,544	0.0	11,922	0.0	7780	0.00	51,754	0.00
	20	86,400	39.3	19,162	0.0	25,453	0.0	12,802	0.00	54,936	0.00
	25	86,400	62.2	35,694	0.0	36,465	0.0	29,719	0.00	61,904	0.00
	30	86,400	58.0	63,452	2.3	34,335	0.0	37,542	0.00	67,092	1.70
30	10	53,806	0.0	28,587	0.0	20,156	0.0	5296	0.00	52,372	0.00
	15	86,400	41.7	45,210	0.0	27,481	0.0	20,331	0.00	60,992	0.00
	20	86,400	79.4	46,737	0.7	38,126	0.0	25,750	0.00	62,452	0.48
	25	86,400	184.3	84,704	2.8	70,184	3.1	78,520	4.03	81,956	3.54
	30	86,400	498.1	82,585	2.6	86,400	5.1	67,884	2.86	83,595	409.97
Average		57,244	46.8	23,784	0.4	20,210	0.3	16,510	0.28	43,197	16.66

In addition, the results show that the VSS is approximately 95% for all instances. This highlights the importance of accounting for stochasticity when modeling the CRP problem, as solutions obtained by solving the “mean-value” problem are not good enough for the stochastic problem. To respect the page limit, we have presented the results of the L-shaped algorithm in online Appendix F. However, it is worth noting that the B&BC algorithm outperformed the L-shaped algorithm when using the same combination of accelerators.

6.4. Marginal impacts of accelerators

To evaluate the impact of individual enhancements, we designate BBC_I as the benchmark and sequentially disable each accelerator, examining their effects on the same instances detailed in Table 4. The summarized results can be found in Table 5. In this tables, under “ $BBC_{I \setminus \{LBF\}}$ ” and “ $BBC_{I \setminus \{PO\}}$ ”, the performance of the best algorithm, i.e., BBC_I , without LBF valid inequalities and PO cuts, is given, respectively. Additionally, we report the results for the BBC_I algorithm without warm-up and BDD under columns “ $BBC_{I \setminus \{Warm-up\}}$ ”, and “ $BBC_{I \setminus \{BDD\}}$ ”, respectively. Column “ $BBC_{I \setminus \{LDD\}}$ ” gives information about BBC_I algorithm when LDD is added to it. Comparison of the results in these figures with the gaps reported for the BBC_I algorithm in Table 4 shows that each accelerator used in the BBC_I algorithm positively marginalizes the gap. Furthermore, we have analyzed the combined effects of various enhancements in detail in Online Appendix E, where the impact of different strategies on solution time, gap reduction, and algorithmic efficiency is thoroughly evaluated.

Additionally, our analysis revealed that among the enhancements to our algorithm, the LBF valid inequalities have the most significant marginal impact on efficiency, whereas the PO cuts have the least. Excluding LBF valid inequalities leads to longer solution times and prevents the algorithm from finding optimal solutions in 15 instances. Notably, within our algorithm, the inclusion of BDDs uniquely extends the average solution time—from 16,510 s without BDDs to 19,193 s when they are incorporated. This increase likely stems from the necessity of solving additional MIP subproblems at each iteration. However, despite the longer solution times, the use of BDDs offers substantial benefits, specifically for large-size instances, by contributing to a reduction in both the optimality gap and the number of nodes explored in the search tree by approximately 2%–3% and 7000 nodes, respectively. To easily compare the performance of the algorithms, we visualize their performance in the performance profile of Fig. 3.

This performance profile reports the computing time in seconds on the horizontal axis and the cumulative percentage of instances being solved to optimality within a given CPU time on the vertical axis. Unlike Table 5, which averages the results of 5 instances per row, Fig. 3 aggregates the outcomes for all 125 instances utilized in evaluating the algorithms. It shows that our benchmark algorithm, BBC_I , outperforms the alternatives.

6.5. Impact of casualty shortage penalty parameters on model behavior

To evaluate the sensitivity of our model to variations in the penalty cost for unmet casualty demand, we conducted a series of computational experiments. Specifically, we examined how changes in the shortage penalty parameters, ρ_{pl}^{high} and ρ_{pl}^{low} , influence optimal decisions and computational performance.

For this analysis, we selected an instance with $|J| = 10$, $|L| = 30$, and $|S| = 100$. The penalty costs ρ_{pl}^{high} and ρ_{pl}^{low} determine the trade-off between minimizing total costs and prioritizing casualty treatment. We systematically varied the penalty cost multipliers, starting from an extremely

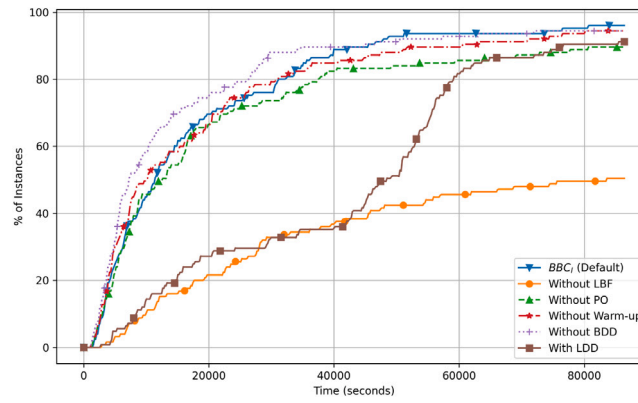
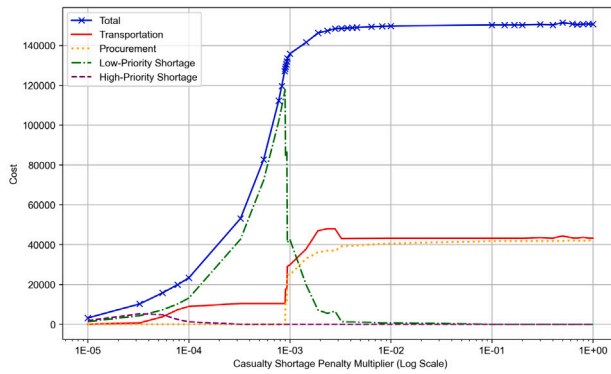
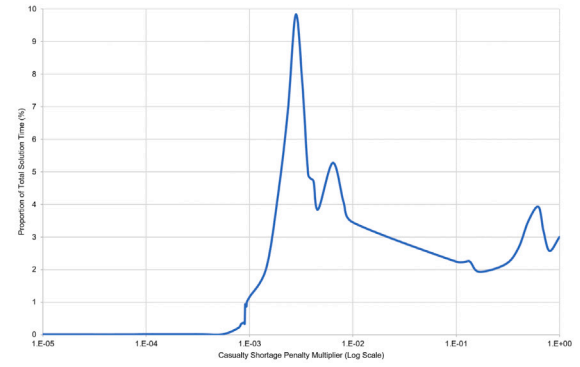


Fig. 3. Percentages of instances solved to optimality within given computing times.



(a) Impact of Shortage Penalty Parameter on Costs.



(b) Impact of Shortage Penalty Parameter on Solution Time.

Fig. 4. Impact of shortage penalty parameter on model behavior.

small value and gradually increasing it up to 1. Throughout this process, we maintained ρ_{pl}^{high} at twice the value of ρ_{pl}^{low} to consistently prioritize high-priority casualties. The objective of this experiment was to assess:

- The impact of increasing penalty costs on total costs, including transportation, procurement, and shortage costs for both casualty priority levels.
- How the model adapts its resource allocation and casualty transportation decisions as the penalty for untreated casualties increases.
- The computational burden imposed by different penalty levels, particularly in terms of solution time.

Fig. 4(a) illustrates how total costs and their components — transportation, procurement, and shortage costs for both priority levels — evolve as a function of the penalty multiplier. Fig. 4(b) presents the proportion of total solution time spent at different penalty levels.

From Fig. 4(a), we observe that when the shortage penalty is set to an extremely small value, the model has no incentive to allocate resources for casualty transportation. Consequently, 100% of casualties remain untreated, leading to the highest possible shortage rate for both high- and low-priority casualties. This is reflected in Fig. 4(b), where solution times are minimal, as the optimization process is straightforward under these conditions.

As the penalty multiplier increases, the cost associated with high-priority casualties rises steadily, though at a decreasing rate, since the model prioritizes their treatment. Low-priority casualties, however, continue to be neglected. A crucial turning point occurs at approximately 10^{-3} , where the model is forced to reassess its allocation strategy. At this stage, we observe a sharp decline in low-priority shortage costs in Fig. 4(a), indicating that the model starts allocating resources to these casualties. Simultaneously, Fig. 4(b) shows a peak in computational effort, suggesting that the model faces its most complex trade-off at this stage.

Beyond this threshold, as the penalty cost continues to increase, the model prioritizes treating all casualties, ensuring no one is left unattended. This results in a minimal shortage rate for both priority levels. Interestingly, solution time slightly decreases beyond this point, as the model transitions from making complex trade-offs to following a more structured and straightforward allocation strategy.

These findings underscore the importance of carefully selecting penalty parameters in casualty response planning. Given the life-and-death nature of post-disaster decisions, an inadequately chosen penalty structure could lead to a large number of untreated casualties, potentially worsening the humanitarian impact of the disaster.

7. Case study

Apart from evaluating the generated instances, we also examined the effectiveness of our proposed algorithm in solving a CRP problem derived from an earthquake case study in Turkey with real data. Our study aimed to identify the location of ACFs in Van Province, Turkey,

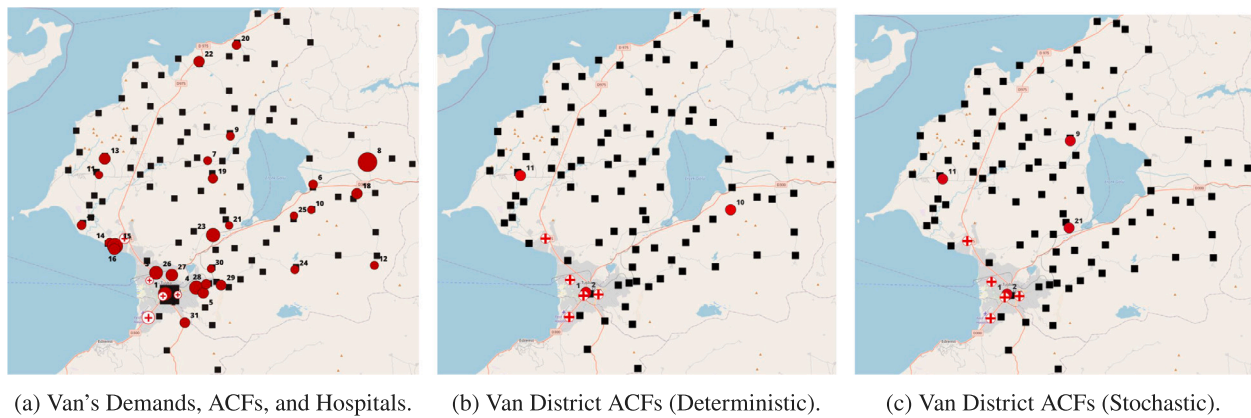


Fig. 5. Comparison of ACF establishments in Van district for deterministic and stochastic models.

following the earthquake of 2011. Van district in Turkey is located in a seismically active region and, as a result, has experienced numerous earthquakes throughout its history. The eastern Turkish province of Van experienced a catastrophic earthquake on October 23, 2011, which measured 7.2 on the Richter scale and resulted in numerous fatalities. A second harmful tremor (5.6 on the Richter scale) struck near Van City two weeks after the initial earthquake in October. These two earthquakes had a profound impact, as over 600 individuals lost their lives, and 2500 suffered injuries (International Federation of Red Cross and Red Crescent Societies, 2012). The 2011 Van earthquake has been the subject of numerous scholarly studies, covering a range of aspects such as last-mile relief network design (Noyan et al., 2016; Noyan and Kahvecioğlu, 2018), humanitarian relief network design (Elçi et al., 2018), shelter location (Kılıcı et al., 2015) and assessment planning (Balcik and Koğlu, 2020). We contribute to this stream of research by studying the location-allocation of ACFs in the Van district after the disaster utilizing a dataset from the related natural disaster. We mainly follow the data scheme presented by Oksuz and Satoglu (2020) and Noyan et al. (2016). Below, we outline the key parameters for the case study.

We have 94 demand neighborhoods affected by the earthquake (squares in Fig. 5(a), with size indicating demand) and 31 potential ACF locations (circles in Fig. 5(a), with size indicating capacity). Noyan et al. (2016) also grouped these 94 points into 30-point and 60-point clusters using a p -median model, which minimizes the total demand weighted travel time. We also evaluate our model on these two versions. Each neighborhood is classified into one of three damage intensity levels, “destructive,” “damaging,” and “strong or very strong,” based on the earthquake’s intensity and distance from the fault line. Each damage intensity level corresponds to different damage state ratios, including “no damage,” “slight and medium damage,” and “heavy damage and collapse” for buildings in the neighborhood. The base demand for each location was then determined by considering the neighborhood’s population, the damage intensity level, the damage state, and the total number of injuries resulting from the 2011 Van earthquake (see the online Appendix in Noyan et al. (2016), for details). To realize demands for each scenario, we multiply the base demands by a deviation factor. These deviation factors are sampled from uniform distributions, which have intervals of [0.75, 1.10], [0.75, 1.20], and [0.75, 1.30] for intensity levels of “destructive,” “damaging,” and “strong or very strong,” respectively.

In the Van district, five primary hospitals are available to assist patients during a disaster (represented by circles with a plus inside in Fig. 5(a), with size indicating capacity). However, in the event of an earthquake, hospital treatment capacities become uncertain. Therefore, we assume their capacity fluctuates between 70% and 120% of their nominal capacity (Peleg and Kellermann, 2009; Oksuz and Satoglu, 2020). We also compute the distance from the geographical coordinates.

The average optimality gap of approximately 0.79% signifies that our proposed algorithm is capable of generating high-quality solutions. Another important criterion for evaluating a stochastic mathematical model is the Value of the Stochastic Solution (VSS). VSS is a standard metric that quantifies the expected benefit of using a stochastic model instead of its deterministic counterpart. It is computed by comparing the solution values of the stochastic and expected-value deterministic versions of the problem (Birge and Louveaux, 2011). A high VSS indicates that explicitly accounting for uncertainty leads to significantly better solutions, justifying the additional computational effort required for stochastic optimization. For a detailed explanation of the formulation and computation of VSS in our study, we refer the reader to Section 6.3.

In our case study, the substantial VSS values, averaging 97.79%, highlight the critical role of stochasticity in disaster management. This indicates that by explicitly accounting for uncertainty and using the stochastic model instead of its deterministic counterpart, overall costs can be reduced by approximately 98% on average. This underscores the importance of incorporating uncertainty into decision-making to enhance response effectiveness. Let us now delve deeper into the solutions. In both deterministic and stochastic models, we established ACFs with specific capacities and the ability to treat certain injuries. In both models, we established the following ACFs:

- ACF #1 (with a capacity of 750, capable of treating the following injuries: fractures/crush (Injury #1), laceration/contusions/open wounds (Injury #2), peripheral nerve injury (Injury #3), and burns (Injury #4)),
- ACF #2 (with a capacity of 570, treating Injuries #2 and #3),
- and ACF #11 (with a capacity of 52, treating injury #2).

However, in the deterministic model, we opened an additional ACF, #10, with a capacity of 70 patients capable of treating injuries #1, #2, and #4 concurrently (Fig. 5(b)). In the stochastic model, considering 100 different scenarios with varying patient numbers and hospital treatment capacities, we established two more ACFs: ACF #9 (treating injuries #1 and #2) and ACF #21 (specializing in treating Injuries #1 and #4), with capacities of 130 and 100, respectively (Fig. 5(c)). Notably, the decision to incorporate two additional ACFs, namely #9 and #21, rather than just one (e.g., ACF #10), results in a total capacity increase of 160 patients compared to the deterministic model. This decision not only allows us to provide care for more patients but also enhances our geographical coverage, leading to reduced patient transport times.

Table 6In-sample and out-of-sample performance of BBC_I with stochastic and fixed hospital caps.

Instance	In-Sample		Out-of-Sample (Trained with Sto. Hos. Treat. Cap.)			Out-of-Sample (Trained with Det. Hos. Treat. Cap.)			$\Delta^{Dem.}$ (%)	
	$ J $	$ L $	Time (s)	Gap (%)	Ave. Obj.	Unsat. Dem. (#)	Unsat. Dem. (%)	Ave. Obj.		Unsat. Dem. (#)
31	30	75,536	0.00	129,935	1	0.01	309,783	3	0.12	66.67
	60	86,400	2.38	595,402	8	0.30	906,262	12	0.48	33.33
	94	80,160	0.00	1,705,770	24	0.93	16,475,700	232	9.27	89.66
Average		80,699	0.79	810,369	11	0.41	5,897,248	82	3	63.22

Examining the challenging scenarios that can arise after an earthquake, particularly when we base decisions on the average demand in the deterministic model, we come across a critical situation in scenario #22, among 100 scenarios generated for our case study. This scenario has the highest number of low-priority cases associated with Injuries #1 and #2. Moreover, if first-stage decisions are made based on the average scenario in the deterministic model, this scenario also leads to the highest number of unsatisfied low-priority patients. In scenario #22, we encounter a remarkable 1596 low-priority patients, with 402 patients falling into class #3, the most heavily populated patient class within this scenario. Upon closer examination of class #3, we discover that it exclusively comprises patients with Injuries #1 and #2. This revelation underscores the necessity of an ACF specializing in treating these two specific injuries, much like ACF #9 in the stochastic model. This level of responsiveness to the unique demands of various scenarios is a notable advantage of the stochastic approach, and it addresses an oversight in the deterministic model where such specialized care was not adequately considered.

Effective collaboration among various organizations is crucial for ensuring the availability of necessary equipment, an often overlooked aspect in post-disaster scenarios. The differences between the deterministic and stochastic models extend beyond the number, capacity, and treatment capability of the established ACFs. For example, the stochastic model significantly increases the number of rescue vehicles of various types assigned to ACFs to ensure all low-priority patients can be attended to in every scenario, including worst-case scenarios. Concerning the number of physicians, while the deterministic model accurately estimated the number of required orthopedic physicians, it significantly underestimated the need for wound specialists and plastic physicians or dermatologists and overestimated the number of required neurologists. This highlights the importance of a collaborative effort among companies and organizations to ensure that the right mix of medical professionals and resources is readily available in post-disaster situations, as the requirements vary considerably between deterministic and stochastic models.

7.1. Evaluating out-of-sample performance and the impact of hospital capacity uncertainty

To assess the robustness of our stochastic model and ensure its real-world applicability, we conducted an out-of-sample evaluation. While our model is trained using 100 in-sample scenarios derived from historical data of the 2011 Van earthquake, it is crucial to test how well it generalizes to previously unseen disaster conditions. This ensures that the first-stage decisions remain effective even when faced with variations in patient demand and hospital treatment capacities beyond those explicitly considered in the optimization model used to determine the first-stage plan.

Our evaluation followed these steps: First, we solved the casualty response planning problem for the Van earthquake using 100 in-sample scenarios, generated based on historical data and expert assessments (as discussed in the introduction of Section 7). Once the model produced an optimal first-stage solution — determining the locations of ACFs and resource allocations — these decisions were fixed. Next, we tested their robustness on 2000 new out-of-sample scenarios, generated from real-world data distributions but not included in the in-sample set.

Since our model penalizes unsatisfied casualties heavily, even a single untreated casualty can significantly impact the objective function, making it an unreliable measure for evaluating solution quality. To provide a more practical assessment, we define a new key performance indicator (KPI): *Percentage of Unsatisfied Demand*, which quantifies the proportion of casualties left without treatment when applying the first-stage solution to out-of-sample scenarios. This metric ensures that our optimized response plan remains effective even under previously unseen disaster conditions. The results are presented in Table 6, under two main columns:

- **Out-of-Sample (Trained with Sto. Hos. Treat. Cap.):** Evaluates performance when the model was trained using stochastic demand and uncertain hospital treatment capacities.
- **Out-of-Sample (Trained with Det. Hos. Treat. Cap.):** Evaluates performance when the model was trained using stochastic demand and fixed hospital treatment capacities.

In this table, the first two columns provide the number of potential locations for the ACFs and the number of demand locations, respectively. “Time (s)” refers to the total solution time. Column “Gap (%)” computes the gap between the lower- and the upper-bound for in-sample scenarios. Additionally, for out-of-sample evaluations under both stochastic and deterministic hospital treatment capacities, we include “Ave. Obj.”, which represents the average objective function value. Columns “Unsat. Dem. (#)” and “Unsat. Dem. (%)” show the number and percentage of casualties left untreated in disaster areas, respectively. Lastly, “ $\Delta^{Dem.}$ (%)” measures the reduction in untreated casualties when accounting for stochastic hospital capacities in the in-sample scenarios.

From Table 6, we observe that the average optimality gap of BBC_I in the case study is 0.79%, demonstrating that our algorithm consistently finds high-quality solutions. More importantly, the out-of-sample evaluation shows that, on average, only 0.41% of total casualties remain unsatisfied when applying the optimized first-stage decisions to scenarios unseen during model training. This exceptionally low percentage suggests that decisions made while explicitly considering both the assessment information provided by key informants and historical disaster data can remain efficient when evaluated using independently generated scenarios designed to assess the impact of uncertainty.

We also examine the role of hospital capacity uncertainty in decision-making. The results highlight a significant difference: When hospital capacity uncertainty is ignored in the in-sample optimization, out-of-sample casualty shortages increase significantly. Specifically, when comparing solutions obtained with and without hospital capacity uncertainty, the percentage of unsatisfied casualties increases from 0.01%, 0.30%, and 0.93% to 0.12%, 0.48%, and 9.27%, respectively. This corresponds to a 63.22% higher number of unsatisfied casualties when hospital capacity uncertainty is neglected. These findings underscore the importance of incorporating hospital capacity uncertainty into casualty response planning models. Failing to account for it leads to significant misallocations and increased casualty shortages in real-world conditions.

8. Conclusion

In this paper, we addressed the challenge of a casualty response planning problem with patients having multiple injuries. To tackle this problem, we developed two two-stage stochastic mixed-integer programming models to formulate the CRP with uncertain demands and hospital treatment capacities. The first model is a basic mathematical model, whereas the second one is a more intricate version that significantly reduces solution times. We proved that the two proposed models are equivalent. As our solution methods, we introduced both the L-shaped algorithm, a pure cutting-plane method tailored to our stochastic mathematical model, and the branch-and-Benders-cut algorithm. In order to significantly improve the efficiency of these algorithms, we have incorporated a diverse set of acceleration techniques, including Benders dual decomposition, Lagrangian dual decomposition, a multi-cut reformulation, the integration of Pareto-optimal cuts, the utilization of lower bounding functional valid inequalities, partially relaxed subproblems, and heuristic cut generation strategies. Our computational results showed that the improved two-stage stochastic programming model is around 42.5% faster than the initial model. We demonstrated that the proposed acceleration techniques significantly improve the performance of the algorithms and reduce the average optimality gap from 563.1% to 0.1%. Particularly, in some instances, the gap improved from nearly 7,900% to around 1%. We also observed that the average VSS is 95.7%, that demonstrates the importance of addressing uncertainty in casualty response planning problem and highlights the advantage of our proposed two-stage stochastic programming model compared to its corresponding deterministic model. We also provided extensive results to measure the contribution of enhancements separately. To highlight the applicability of our proposed models and algorithms, we ran our best algorithm on a case study from Turkey and demonstrated that we can find near-optimal solution with an average gap of 0.8%. Additionally, we benchmarked our best-performing algorithm, BBC_I , against PHA, a widely used decomposition approach in disaster response operations. The results showed that BBC_I improved the objective function by 90.76% over PHA while requiring significantly less computation time, demonstrating its superior efficiency and solution quality.

We suggest four important directions for future research. First, given the significant impact of road conditions and travel times in post-disaster scenarios, incorporating debris clearance as an integral component of the problem discussed in this paper would be invaluable. Second, considering secondary disasters within the context of CRP, which includes uncertain demands, stochastic hospital treatment capacities, and multiple injuries, would be interesting. It is important to understand the uncertainty and severity of these secondary disasters and assess their likelihood of affecting hospitals. Third, expanding the model to incorporate uncertainty in both travel times and service times could significantly enhance the realism and effectiveness of post-disaster response planning. While our current work focuses on treatment capacity uncertainty, future research could explore how fluctuating travel conditions and dynamic service times impact resource allocation and patient care itineraries. Such considerations are especially relevant in scenarios where road accessibility changes due to debris accumulation and infrastructure failures. Fourth, exploring dynamic resource management in ACFs, where treatment capabilities adapt to real-time demand and resource availability, could enhance flexibility in medical equipment and personnel deployment. This would improve responsiveness to uncertainty and optimize casualty care in disaster scenarios.

CRedit authorship contribution statement

Pedram Farghadani-Chaharsooghi: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hossein Hashemi Doulabi:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Data curation, Conceptualization. **Walter Rei:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Michel Gendreau:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Data curation, Conceptualization.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.cor.2025.107165>.

Data availability

Data will be made available on request.

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